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INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT
ON A
PROPOSED CREDIT

IN THE AMOUNT OF EUR 92.2 MILLION
(US\$100 MILLION EQUIVALENT)

TO THE

REPUBLIC OF MALI

FOR A

MALI WATER SECURITY PROJECT
(P181538)
DECEMBER 23, 2024

Water Global Practice
Western and Central Africa Region

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CURRENCY EQUIVALENTS

Exchange Rate Effective October 31, 2024

Currency Unit =	WEST AFRICAN FRANC (XOF)
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US\$1 =	EUR 0.92017
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US\$1 =	XOF 603.59
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FISCAL YEAR

January 1 - December 31

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ABBREVIATIONS AND ACRONYMS

ABN	Niger Basin Authority (Autorité du Bassin du Niger)
CERC	Contingency Emergency Response Component
DGDP	General Directorate of Public Debt (Direction Générale de Dette Publique)
DMA	District Metering Areas
DNH	National Directorate of Hydraulics (<i>Direction Nationale de l'Hydraulique</i>)
DNPD	National Directorate of Development Planning (Direction Nationale de la Planification du Développement)
E&S	Environmental and social
ECOWAS	Economic Community of West African States
ESCP	Environmental and Social Commitment Plan
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GRM	Grievance Redress Mechanism
GPN	General Procurement Notice
IDA	International Development Association
IFR	Interim Financial Report
M&E	Monitoring & Evaluation
NBS	Nature-based Solutions
NPV	Net Present Value
NRW	Non-revenue Water
PASEMa	Mali Water Security Project (<i>Projet d'Appui à la Sécurité de l'Eau au Mali</i>)
PDO	Project Development Objective
PEMU	Mali Urban Water Supply Project
PIM	Project Implementation Manual
PMIA	Project Management and Implementation Agency
PPSD	Project Procurement Strategy for Development
PRUBA	Bamako Urban Resilience Project
PSC	Project Steering Committee
SOMAGEP-SA	Malian Drinking Water Management Company (<i>Société Malienne de Gestion de l'Eau Potable</i>)
SOMAPEP-SA	Malian Water Asset Holding Company (<i>Société Malienne de Patrimoine de l'Eau Potable</i>)
STEP	Systematic Tracking and Evaluation Process
WaterSAP	Water Sector Assessment



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**DATASHEET****BASIC INFORMATION**

Project Beneficiary(ies)	Operation Name		
Mali	Mali Water Security Project		
Operation ID	Financing Instrument	Environmental and Social Risk Classification	
P181538	Investment Project Financing (IPF)	Substantial	

Financing & Implementation Modalities

☐ Multiphase Programmatic Approach (MPA)	☒ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☒ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternative Procurement Arrangements (APA)	☐ Hands-on Expanded Implementation Support (HEIS)

Expected Approval Date	Expected Closing Date
22-Jan-2025	31-Dec-2029
Bank/IFC Collaboration	
No	

Proposed Development Objective(s)

Increase access to improved drinking water supply and enhance conservation of water sources in selected cities.

Components

Component Name	Cost (US\$)
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Component 1: Water supply in Bamako and selected secondary centers	87,000,000.00
Component 2: Enhancing the conservation of the water sources	8,000,000.00
Component 3: Project management and Capacity reinforcement	5,000,000.00
Component 4: Contingency Emergency Response Component	0.00

Organizations

Borrower:	Republic of Mali		
Contact	Title	Telephone No.	Email
Djoouro Bocoum	Secretary General	22376023156	bdjoouro@yahoo.fr
Implementing Agency:	Direction National de l'Hydraulique (DNH), Societe Malienne de Gestion de l'Eau Potable (SOMAGEP), Societe Malienne de Patrimoine de l'eau Potable (SOMAPEP)		
Contact	Title	Telephone No.	Email
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PROJECT FINANCING DATA (US\$, Millions)

Maximizing Finance for Development

Is this an MFD-Enabling Project (MFD-EP)? Yes

Is this project Private Capital Enabling (PCE)? No

SUMMARY

Total Operation Cost	100.00
Total Financing	100.00



of which IBRD/IDA	100.00
Financing Gap	0.00

DETAILS**World Bank Group Financing**

International Development Association (IDA)	100.00
IDA Credit	50.00
IDA Shorter Maturity Loan (SML)	50.00

IDA Resources (US\$, Millions)

	Credit Amount	Grant Amount	SML Amount	Guarantee Amount	Total Amount
National Performance-Based Allocations (PBA)	50.00	0.00	50.00	0.00	100.00
Total	50.00	0.00	50.00	0.00	100.00

Expected Disbursements (US\$, Millions)

WB Fiscal Year	2025	2026	2027	2028	2029	2030
Annual	1.00	5.00	20.00	45.00	25.00	4.00
Cumulative	1.00	6.00	26.00	71.00	96.00	100.00

PRACTICE AREA(S)**Practice Area (Lead)**

Water

Contributing Practice Areas

Climate Change

CLIMATE



Climate Change and Disaster Screening

Yes, it has been screened and the results are discussed in the Operation Document

SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category	Rating
1. Political and Governance	● High
2. Macroeconomic	● Substantial
3. Sector Strategies and Policies	● Substantial
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Moderate
6. Fiduciary	● Substantial
7. Environment and Social	● Substantial
8. Stakeholders	● Substantial
9. Overall	● Substantial

POLICY COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

ENVIRONMENTAL AND SOCIAL

Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

E & S Standards	Relevance
ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Relevant



ESS 10: Stakeholder Engagement and Information Disclosure	Relevant
ESS 2: Labor and Working Conditions	Relevant
ESS 3: Resource Efficiency and Pollution Prevention and Management	Relevant
ESS 4: Community Health and Safety	Relevant
ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Relevant
ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Relevant
ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Not Currently Relevant
ESS 8: Cultural Heritage	Relevant
ESS 9: Financial Intermediaries	Not Currently Relevant

NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).

LEGAL

Legal Covenants

Sections and Description

Schedule 2. Section I.A.2. (a): The Recipient shall establish not later than three (3) months after the Effective Date, or such later date as agreed by the Association, and maintain throughout Project implementation, a steering committee for the Project, with a mandate, composition and resources acceptable to the Association ("Project Steering Committee" or "PSC").

Schedule 2. Section I.B.2. (c): The obligation of the Project Implementing Entity to maintain, throughout Project implementation adequate staffing, personnel and procedures for the implementation of the Project, as further detailed in the Project Implementation Manual, including, inter alia, (i): the recruitment of an external auditor not later than six (6) months after the Effective Date; and (ii): the updating of the accounting software not later than three (3) months after the Effective Date; or such later date as agreed by the Association.

PA. Schedule. Section I.A.2.a.(i): To facilitate the carrying out of Part 1.3 of the Project, the Project Implementing Entity shall not later than one (1) month after the Effective Date or any later date as agreed by the Association, enter into a convention with SOMAGEP-SA ("Convention SOMAGEP-SA"), under terms and conditions acceptable to the Association in accordance with the Project Implementation Manual.

PA. Schedule. Section I.A.2.b.(i): To facilitate the carrying out of Part 2 of the Project, the Project Implementing Entity shall not later than one (1) month after the Effective Date, or any later date as agreed by the Association, enter into a convention with DNH ("Convention-DNH") under terms and conditions acceptable to the Association and in accordance with the Project Implementation Manual.

PA. Schedule. Section I.B.2.(c): The Subsidiary Agreement shall include, inter alia, the following provisions: the obligation of the Project Implementing Entity to maintain, throughout Project implementation adequate staffing, personnel and procedures for the implementation of the Project, as further detailed in the Project Implementation



Manual, including, inter alia, (i) the recruitment of an external auditor not later than six (6) months after the Effective Date; and (ii) the updating of the accounting software not later than three (3) months after the Effective Date; or such later date as agreed by the Association.

ESCP: Recruit or appoint an environmental specialist, a social issues specialist and four (04) junior environmental and social safeguards specialists no later than 30 days after effectiveness, then maintain these positions throughout the implementation of the project.

ESCP: Establish the complaints management mechanism 60 days after the project effectiveness, and then maintain and operate this mechanism throughout the implementation of the project.

Schedule 2,Section I.B.1: To facilitate the carrying out of the Project, the Recipient shall make the proceeds of the Financing available to the Project Implementing Entity (“Subsidiary Financing”) under a subsidiary agreement between the Recipient and the Project Implementing Entity, under terms and conditions approved by the Association (“Subsidiary Agreement”).

Conditions

Type	Citation	Description	Financing Source
Effectiveness	5.01.a	The Recipient has caused the Project Implementing Entity to prepare and adopt the Project Implementation Manual in form and substance acceptable to the Association.	IBRD/IDA
Effectiveness	5.01.b	The Subsidiary Agreement between the Recipient and the Project Implementing Entity has been executed and delivered, and all conditions precedent to its effectiveness or the right of the Project Implementing Entity to make withdrawals under it (other than the effectiveness of this Agreement) have been fulfilled.	IBRD/IDA



I. STRATEGIC CONTEXT

Introduction

1. **Mali faces multifaceted problems that are delaying its economic development including water insecurity, aggravated by climate change, natural resource degradation, fragility, and rapid population growth.** The country is highly vulnerable to climate change impacts, such as increasing severity and incidence of floods and droughts, that decrease food security and limit agricultural production, upon which most of the population depends. Poor water resource management, aggravated by limited development of these water resources, further impact already fragile livelihoods, worsening economic and human capital outcomes. Poor coordination among institutions responsible for management and use of natural resources, including water, increases the country's vulnerability to climate change impacts.

2. **Well-coordinated, integrated water-related interventions are needed to increase resilience to climate variability, reduce fragility, and strengthen the foundations of socioeconomic development in Mali.** The proposed project is designed to enhance climate resilience by increasing water security in selected cities and their corresponding sub-basins. The main goal of the project is to enhance resilience to climate change—exacerbated drought risk by helping the government increase sustainable access to drinking water services in an extension of the Kabala program¹ in Bamako and secondary centers while securing water resources in the corresponding sub-basins through activities implemented by local community organizations. Through harmonization across the World Bank portfolio and with other partners, a coordinated approach to water security will encompass water resource management, ecosystem preservation, irrigation, optimization of rain-fed agriculture, flood management, and access to properly managed sanitation.

A. Project Strategic Context

3. **Mali faces internal and external shocks that hamper progress toward inclusive, sustainable development.** Its Human Development Index is low, because of the weakness of its education, health care, and social protection systems. Climate change, rapid demographic growth, and insecurity have increased its vulnerability and fragility. A vast landlocked country of 22.4 million people, Mali has an average annual population growth of 3.3 percent (2009-2022) and a median population age of 21.4². Most of the population lives in the south, close to the Senegal and Niger rivers, with 57 percent in rural areas and 12 percent in the capital city of Bamako. The latest estimates for 2023 suggest that Mali's gross domestic product (GDP) per capita is US\$897 (current prices). Forty-five percent of the Malian population was living under the national poverty line in 2021/22, concentrated in rural areas.³ Access to basic services and infrastructure is limited and highly unequal, with extremely marked regional disparities. In 2021/22, poverty rates were highest in Sikasso (57.1 percent), where households depend on cotton farming, and high in Ségou (53.2 percent) and Mopti (57.5 percent), where families mostly depend on rainfed sorghum and millet cultivation. In Kayes, where households depend on remittances and livestock, poverty is lower (41.4 percent), and the lowest poverty rates are in Bamako (15.8 percent) and Kidal (4.9 percent).⁴ Mali ranks 153 out of 166 countries on the Gender Inequality Index, reflecting significant challenges in reproductive health, empowerment, and economic status of women.⁵

¹ The Kabala program, which the government launched in 2013 with the support of seven donors, is designed to increase water production and expand services in unserved neighborhoods of Bamako.

² RGPH5-2022: Rapport provisoire novembre 2023 – (Fifth General Census of Population and Housing 2022).

³ West African Economic and Monetary Union household survey 2021/22.

⁴ For this entire paragraph, see United Nations estimate (July 2021). According to the results of the Agricultural Economic Survey integrated into household living conditions, the agricultural population stands at 16.8 million, of which 48.3 percent are women (EAC 2018); World Bank (2021) From Covid-19 Crisis response to resilient recovery. Saving lives and Livelihoods while Supporting Green, Resilient and Inclusive Development (GRID). Development Committee of the Boards of Governors of the Bank (The World Bank and IMF) and World Bank (2015) Geography of Poverty in Mali.

⁵ See the Gender Inequality Index (database), United Nations Development Program (accessed November 2021), <http://hdr.undp.org/en/content/gender-inequality->



4. **Mali is among the world's most vulnerable countries to climate change**—ranked 180 out of 187 countries in 2022 on the Notre Dame Global Adaptation Initiative Country Index,⁶ and there has been little structural change in the economy over the last three decades to increase resilience to climate shocks. Most climate scenarios project that temperatures in the Sahel will be at least 2°C higher in 2040 than in 2021, and it is projected that rainfall patterns will become more irregular, with oscillations between very wet and very dry years. Droughts and floods are likely to become more extreme.⁷ More severe droughts will decrease food security in economy reliant on agriculture and put water utilities under stress as water sources dry up. More-extreme floods are likely to damage critical infrastructure, including agricultural and urban water supply networks. Mali has already experienced the consequences of climate change, with major floods in the Senegal and Niger rivers in 2020 affecting 80,000 individuals, 6,000 houses, and 7,000 tons of food.⁸ In at least one project area, the San district, monthly precipitation decreased significantly between 1989 and 2019.⁹ In 2024, severe flooding across the country caused 30 fatalities, affected 47,374 people, and led the government to declare a state of national disaster in August.

5. **Such climatic shocks hamper efforts to reduce poverty by limiting economic growth, especially in the agricultural sector, compounding political instability and insecurity.** Agriculture and low-productivity services dominate the economic and employment landscape, whereas manufacturing is limited and concentrated in agro-industries and cotton ginning. Gold and cotton dominate exports, exposing the economy to commodity-price and climatic shocks. Two-thirds of the country is arid and semiarid and exposed to desertification, and overexploitation and degradation of natural resources has increased vulnerability to climate change. With only 14 percent of land suitable for agriculture, landscape restoration to boost drought resilience is crucial. It is projected that, unless urgent investments in climate adaptation are made, Mali's annual GDP will have shrunk by roughly 10.7 percent by 2050, with 2.2 million additional people in poverty, mainly because of shocks related to climate change.¹⁰

B. Sectoral and Institutional Context

Water resource challenges

6. **Mali receives an estimated 415 billion m³ of precipitation per year, with renewable water from the Niger, Senegal, and Volta River basins accounting for about 71.4 billion m³ per year.**¹¹ Four international river basins shape the country's hydrology: the Niger, Senegal, Volta, and Comoé-Bia-Tanoé. Mali also contains two major transboundary aquifer systems: the Taoudeni Basin (or Taoudeni-Tanezrouft Aquifer System) and the Iullemeden Aquifer System. Temporal and geographic water variability, exacerbated by changing rainfall patterns, contributes to economic water scarcity. With annual renewable water resources of 6,290 m³ per capita, Mali's agricultural water consumption (4.23% of renewable water) does not exceed sustainable water resources at the national level. However, extreme variations in water availability between regions and seasons, exacerbated by climate change and inadequate management, lead to glaring inequalities in access to water, exacerbating water insecurity. Insecurity has restricted movement, disrupted economic activities and limited access to fields and pastures, leading to the abandonment of arable land.

index-gii.

⁶ Notre Dame Global Adaptation Initiative Country Index (database) (accessed 2022) <https://gain.nd.edu/our-work/country-index/rankings/>

⁷ German Federal Ministry for Economic Cooperation and Development. "Climate Risk Profile: Mali." https://www.pik-potsdam.de/en/institute/departments/climate-resilience/projects/project-pages/agrica/climate-risk-profile_mali_en.

⁸ Marc Flessa. 2021. "What Is the Role of Climate Change as a Conflict Multiplier in Mali?" University College of Friesland, Groningen, the Netherlands. <https://campus-friesland.studenttheses.ub.rug.nl/72/1/BA%203785351%20M%20Flessa.pdf>.

⁹ Diarra Doukoro, Georges Abbevi Abby, and Traore Kalifa. 2022. "Drought Monitoring and Assessment of Climate Parameters Variability in Koutiala and San Districts, Mali." *American Journal of Climate Change* 11 (3). <https://www.scirp.org/journal/paperinformation?paperid=120189>

¹⁰ World Bank. 2022. *Sahel Region: Country Climate and Development Report*. Washington, DC: World Bank Group.

¹¹ Ministry of Energy, Water and Mining. 2006. "Annual Report on Water-Mining Resources." Ministry of Energy, Water and Mining, Bamako, Mali.



7. **Climate change and human activities are also degrading the quality of surface waters, particularly through increased runoff and sediment transport..** Changes in rainfall patterns, gold mining in riverbeds, deforestation, soil erosion, and pollution from point and non-point sources are degrading raw water quality, significantly increasing water treatment costs in cities such as Bamako.¹² Lack of knowledge about and limited monitoring of the country's water resources hinder sound decision-making regarding sustainable, climate-resilient water investments.

Water supply and sanitation

8. **Although access to drinking water continues to increase, overall service quality remains poor.** Eighty-three percent of the population has access to water (94 percent in urban areas, 74 percent in rural areas), but only 62 percent of services qualify as improved (JMP 2022)¹³. Because of years of climate shocks, compounded by poor water service management (e.g., underinvestment, failure to anticipate high population growth, poor service provider performance in rural areas), water supply services fail to deliver an adequate amount of good-quality water. In addition, insecurity makes it difficult to develop basic services such as water and electricity. Water supply is the responsibility of the Ministry of Energy and Water (Ministère de l'Énergie et de l'Eau-MEE), which delegates urban asset management to the Malian Potable Water asset holding Company (*Société Malienne de Patrimoine de l'Eau Potable*, SOMAPEP-SA) and the Malian Drinking Water Management Company (*Société Malienne de Gestion de l'Eau Potable*, SOMAGEP-SA), whereas rural water is the responsibility of the National Directorate of Hydraulics (*Direction Nationale de l'Hydraulique*, DNH) and municipalities. The urban water perimeter covers 90 centers but lacks the resources to upgrade their systems. With 65 percent water coverage,¹⁴ several neighborhoods in Bamako experience water shortages that an increase in demand and drought, heightened by climate change, could exacerbate. Shortages due to low pressure and limited water network are acute in the oldest neighborhoods. To address these challenges, the government launched the Kabala program in 2013 with seven donors to increase water production and expand services in unserved neighborhoods of Bamako, but incomplete funding limits service delivery.

9. **People who have been internally displaced because of climate shocks and insecurity add pressure to already-inadequate water and sanitation services.** Mali has approximately 400,000 internally displaced persons, with many residing in project areas such as Mopti. Approximately 28 percent of the population in these areas lacks access to basic water and sanitation services, underscoring a critical need. The needs of displaced people and the pressure on existing facilities, are not being met by the government's response, which may not be adequate.

10. **The financial performance of the water sector is weak.** Expansion of the urban sector zone in 2018 to include 72 additional secondary centers with low coverage (31 percent, compared with 64 percent in the original 18 centers),¹⁵ has hindered the country's ability to meet the Sustainable Development Goals and to charge adequate tariffs for services,¹⁶ exacerbating the sector's financial situation. Mali's water tariffs, which have remained unchanged since 2004, are among the lowest in West Africa, are insufficient to cover operating expenses, and do not allow for necessary investments in climate resilience. In 2020, a World Bank-funded study on the financial viability¹⁷ of the sector recommended debt relief in the short term to stabilize the sector, and the introduction of structural policies in the medium and long term, such as (i) socially acceptable tariff adjustments; (ii) subsidy compensation; and (iii) payment of the reform costs promised to SOMAPEP-SA when the assets of the former Eau et Énergie du Mali were separated.. SOMAPEP's precarious financial situation and level of debt is a major constraint on increasing urban service coverage and improving performance.

¹² Various annual reports from Société Malienne de Gestion de l'Eau Potable.

¹³ JMP 2022 : World Health Organization and UNICEF joint Monitoring Program <https://washdata.org/data/household#!dashboard/new>

¹⁴ Elaboration d'un Plan pour la Viabilité Financière du Secteur de l'Eau Potable, 2020. SOMAPEP-SA/Banque Mondiale.

¹⁵ Elaboration d'un Plan pour la Viabilité Financière du Secteur de l'Eau Potable, 2020. SOMAPEP-SA/Banque Mondiale.

¹⁶ Under the now-closed Mali Urban Water Supply Project (P122826), the Bank has financed feasibility studies for improvement of the water supply in 20 secondary towns that SOMAPEP-SA and SOMAGEP-SA serve.

¹⁷ Elaboration d'un Plan pour la Viabilité Financière du Secteur de l'Eau Potable, 2020. SOMAPEP-SA/Banque Mondiale.



Although SOMAGEP-SA's service delivery performance is generally satisfactory, insufficient funding and lack of commercial viability of secondary centers it manages hinder its ability to provide climate-resilient water services.

11. **Non-revenue water (NRW) and energy costs pose significant challenge to the water sector's financial sustainability and contribute to increased greenhouse gas (GHG) emissions.** With current network efficiency at less than 62 percent, which is considered very poor compared to the efficient range of 80 to 85 percent, there is a need for more water abstraction, which in turn further exacerbates the drought crisis. The financial model developed to monitor the sector's financial viability identified tariff adjustment as a way to increase water usage efficiency and enhance the financial sustainability of the water utility sector but these measures have not been implemented by the government. According to SOMAGEP-SA, the main reasons behind the increase in NRW between 2018 to 2024 were (i) the commissioning of the new Bamako (Kabala) Water Treatment Plant, which puts pressure on the older network, causing numerous water leaks; (ii) water production that generates no revenue in towns experiencing insecurity (e.g., Kidal); and (iii) poor network quality in the 72 secondary centers added to the urban perimeter in 2018. SOMAGEP-SA has a dedicated NRW department with staff who know the network well and are familiar with strategies for detecting leaks, metering, and establishing district metering areas. The project will support the reduction of NRW through a program of network rehabilitation, sectorization, and installation of production and distribution meters.

12. **Only 20 percent of households nationally have access to safely managed sanitation services.**¹⁸ Access to sanitation has progressed at a much slower pace than access to water and is lower than in many countries in the region. Through the Bamako Urban Resilience Project (PRUBA, P171658),¹⁹ the World Bank is developing two major fecal sludge treatment plants, 800 sanitary blocs, and 20,000 household latrines. Implementation of these investments will generate lessons to define future sanitation sector engagement.

II. PROJECT DESCRIPTION

13. Given vulnerability to climate change and other water security challenges that Mali faces, an integrated approach is crucial to sustain economic development and reduce fragility. The proposed project will enhance drought and flood resilience by supporting sector development through a comprehensive water-management approach focused on increasing access to drinking water as a continuation of the Kabala investment program in Bamako and selected cities while protecting associated water resources at the sub-basin level. The project will support important climate-resilient water investments for water supply, water source conservation, and operational and institutional measures to enhance sector performance such as reducing NRW and priority reforms. The project will not only address immediate water access and quality concerns, but also lay the groundwork for long-term resilience and contribute to water-related job growth and economic activity. In supporting these activities, the project will help build a stable, sustainable future for the country; reduce the risks of water-related conflicts; and support economic growth in vulnerable regions.

A. Project Development Objective

The project development objectives are to increase access to improved drinking water supply and enhance conservation of water sources in selected cities.

¹⁸ World Health Organization. 2021. "Progress on Household Drinking Water, Sanitation and Hygiene 2000-2020: Five Years into the Sustainable Development Goals." Geneva: World Health Organization and United Nations Children's Fund.

¹⁹ 250 million IDA aims at increasing access to urban waste, sanitation, and water services, increasing resilience to floods in selected vulnerable areas of Bamako and targeted neighboring communes, and increasing urban management capacity.



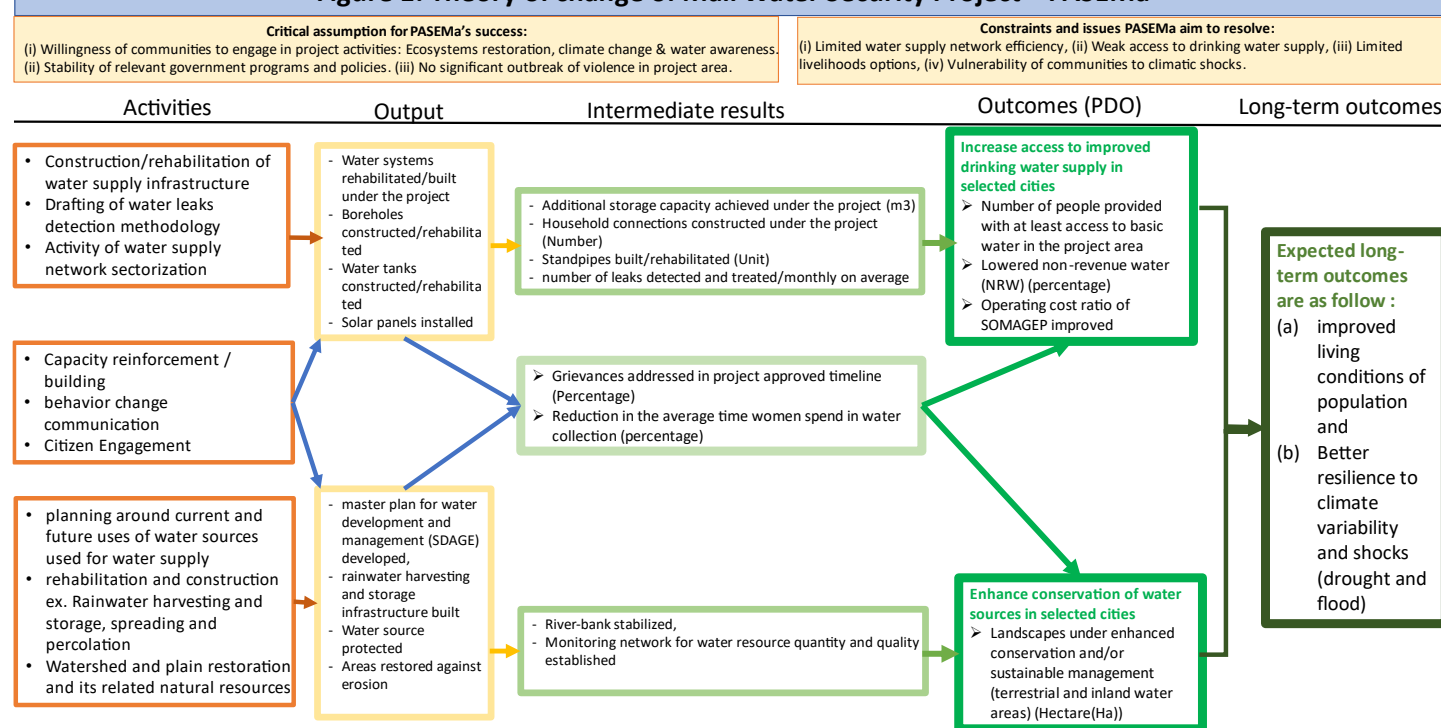
Table 1: Project Development Objectives and Indicators

PDO outcomes	Outcome Indicators
Increase access to improved drinking water supply in selected cities	➤ Lowered non-revenue water (NRW) (percentage) ➤ People provided with at least basic water (Number of people) (corporate scorecard). ➤ Operating cost ratio of SOMAGEP improved (percentage)
Enhance conservation of water sources in selected cities	➤ Landscapes under enhanced conservation and/or sustainable management (terrestrial and inland water areas) (Hectare(Ha)) (corporate scorecard).

B. Theory of Change and PDO Indicators

14. The project results chain is presented in Figure 1 (Theory of Change). In response to mounting pressures on water resources—accelerated by climate change—the project is designed to support the government’s urban water supply program in Bamako and secondary centers, enhance conservation of water sources and increase the capacity of institutions such as SOMAPEP-SA, SOMAGEP-SA, DNH, and local water committees to deliver the expected outcomes of the project. Soft and hard investments under the project are linked to intermediate results, which in turn contribute to achievement of PDO outcomes.

Figure 1: Theory of change of Mali Water Security Project - PASEMa





C. Project Beneficiaries

15. **The main project beneficiaries are households in the project area (Bamako, Bafoulabé/Mahina, Dioila, Mopti/Sévaré, Ouélessébougou, San) and rural populations benefiting from ecosystems restored through nature-based solutions (NBSs).** It is estimated that at least 497,000 people will benefit directly from project investments, approximately half of whom are women and 19.2 percent youth. Categories of beneficiaries include:

- 227,000 people with access to at least basic water services
- 270,000 people who will benefit from improved service quality
- 30,000 people in rural areas near project implementation cities who will benefit from ecosystem restoration

D. Project Components

16. **The proposed project is structured around four components contributing to water security.** Project locations were selected using a rigorous process with key criteria such as maximizing the number of people to be provided with access, as well as focusing on cities vulnerable to climate change (drought, floods), level of water service, readiness of technical studies, and security situation. Most of selected activities have advanced detailed studies. For secondary centers, alternative centers are identified to anticipate possible security challenges that could hinder implementation. The project is described in detail in Annex 3.

17. **Component 1: Water supply in Bamako and selected secondary centers (US\$87 million equivalent):** This component will ensure safe access to drinking water in the face of climate change—exacerbated droughts and floods, by supporting development of efficient, well-performing drinking water services in Bamako and selected secondary towns. The component will leverage technical studies already conducted with the support of the French Development Agency (Agence Française de Développement) in Bamako and secondary centers of Mopti, San, and Sévaré. Special attention will be paid to the most vulnerable and underserved neighborhoods on the left bank of the Niger River in Bamako, which are also most exposed to floods and drought. The component will consist of water supply systems in secondary centers (US\$47 million), a water supply system in Bamako (US\$28 million), and management of NRW (US\$12 million).

18. **Subcomponent 1.1: Water supply systems in secondary centers (US\$47 million):** This subcomponent will construct water supply systems to enhance drought resilience by increasing access to water supply services in selected secondary urban centers through the construction of water supply systems in three selected secondary centers: Mopti, San, and Sévaré. The works to be carried out include construction of water supply systems in secondary urban centers including (e.g., boreholes for production of 24,400 m³/day, tree pumping stations with mixed energy (solar/grid), four 2,000-m³ elevated water tanks and one 1,000-m³ ground-level water tank, water distribution networks, water transmission pipes, social household connections, community standposts) and implementation of protection zones in compliance with national legal requirements upstream of the water production boreholes in the targeted centers.

19. **Subcomponent 1.2: Water supply systems in Bamako (US\$28 million):** This subcomponent will increase drought resilience by constructing and supervising water supply systems in Bamako, mainly a 9-km transfer from Djikoron Para to Missira, a pumping station at Missira, a 6-km transfer from Missira to Kati-Sud, a 2,000 m³ elevated water tank, the distribution network, and household connections.



20. **Subcomponent 1.3: Management of non-revenue water (NRW (US\$12 million)):** This subcomponent will enhance drought resilience by improving the quality of water supply services and reducing energy use by reducing network loss, improving metering for households and large consumers (companies, administration), procuring equipment to ease leak detection, replacing poor segments of the network, increasing capacity for network mapping, and establishing district metering areas (DMAs). Activities under this subcomponent include rehabilitation of networks to replace 100 km of poor segments in Bafoulabé/Mahina, Bamako, Dioila, and Ouélessébougou; establishing district metering areas in Bamako for efficient metering and water supply network management; renewing household and distribution meters for Bafoulabé/Mahina, Bamako, Dioila, Ouélessébougou; installing production meters in Bafoulabé/Mahina, Dioila, and Ouélessébougou; providing technical assistance to SOMAPEP-SA and SOMAGEP-SA to improve technical and commercial performance; procuring equipment to ease leak detection building; and building capacity for network mapping and reinforcement. Such measures will reduce operating costs (a 5 percent reduction in water loss generates 24,255 m³/day of water); increase revenue (and thereby increase financial viability); and reduce the amount of water abstracted from water sources, thus increasing drought resilience.

21. **Component 2: Enhancing conservation of water sources (US\$8 million equivalent):** Infrastructure investments in Component 1 are closely tied to the water resources they depend on and are affected by climate change. To ensure a climate-resilient water supply, this component will support (i) climate-resilient planning of current and future uses of water sources connected to infrastructure investments under Component 1 of the project; (ii) provide DNH and other institutions with data and tools to manage water resources securely (including strengthening the monitoring network for water resource quantity and quality [e.g., at raw water production sites] and effectively communicating monitoring results); (iii) reinforce water allocation tools on the Niger River in collaboration with the Niger Basin Authority to ensure effective management and equitable distribution of vital water resources; (iv) evaluate basin-level dynamics (including turbidity) of water from the Niger and Senegal rivers that is tapped for the selected secondary centers, including through dialogue and coordination with key stakeholders; (v) implement NBSs in collaboration with local communities to support soil and riverbank stabilization in selected areas; (vi) establish and implement immediate protection and close protection areas; and (vii) restore landscapes to preserve and enhance the natural recharge of identified aquifers and the application of NBSs to contribute to climate change mitigation.

22. **Component 3: Project management and capacity reinforcement (US\$5 million equivalent):** This component will support:

- Project management activities, including coordination and supervision of implementation; financial management, procurement, and audits; staffing and training; day-to-day project activities; environmental and social (E&S) safeguard management, including supervision of implementation of an E&S commitment plan and relevant site-specific resettlement action plans as applicable, and operation of the Grievance Redress Mechanism (GRM); monitoring and evaluation (M&E); and citizen engagement activities.
- Capacity-building activities, including in relation to emergency response plans and increasing the climate resilience of the water supply; implementation support, including field missions and data collection; outreach and communications related to the role of the recipient and the project implementing entity; studies of the water management sector; and capacity reinforcement of SOMAPEP-SA, SOMAGEP-SA, and DNH, including those relevant for the water sector assessment (WaterSAP) ²⁰.

²⁰ In parallel with this operation, the World Bank is conducting a water sector assessment to provide recommendations for improving the sector's performance.



23. **Component 4: Contingency Emergency Response Component (CERC) (US\$0 million):** A CERC is included in the project in accordance with Investment Project Financing (IPF) Policy, paragraphs 12 and 13, for Situations of Urgent Need of Assistance and Capacity Constraints. This will allow for rapid reallocation of grant uncommitted funds in the event of an eligible emergency as defined in OP 8.00. A CERC Manual will guide the activation and implementation of the CERC, and an Emergency Action Plan will be prepared to confirm activities and financing for a specific event. Only IDA funds can be reallocated to the CERC.

E. Role of Partners

24. **The World Bank has been a steady partner of the Mali water sector since 2013, supporting significant improvements in sector performance that the proposed project will sustain at a critical juncture.** The World Bank has been supporting Mali in the water sector since 2007 through the Niger Basin Authority (Autorite du Bassin du Niger -ABN).²¹ Since 2011, the World Bank has supported the government of Mali in urban water sector reform, including creation of SOMAPEP-SA (asset holding company) and SOMAGEP-SA (water operator utility). The World Bank has also supported the investment program associated with this reform with the US\$130 million Mali Urban Water Supply Project (PEMU, P122826) implemented from 2013 to 2022. Currently, the water sector in Bamako is being supported through the PRUBA project (P171658), an urban development project with a water and sanitation component. World Bank support in the water sector spans water resource management, ecosystem restoration, irrigation,²² drinking water supply, and sanitation. The World Bank has also supported analytical work since 2012²³ to advance knowledge in the sector. World Bank interventions were conducted in partnership with many donors, including the African Development Bank, European Investment Bank, and French Development Agency (Agence Francaise de Developpement-AFD).

F. Lessons Learned and Reflected in the Project Design

25. **Need for flexibility due to security challenges.** Project implementation in Mali requires state presence and flexibility, for example through the possible change of intervention areas and in implementing activities through nongovernmental organizations or communities in the case of an increase in insecurity. The project will intervene in some areas affected by insecurity (Mopti and San) to indicate the presence of the state. The project also has flexibility to intervene in alternative secondary cities without restructuring if the security situation deteriorates in the selected areas.

26. **Water supply services should be developed in response to demand.** During implementation of previous project (PEMU), it was difficult to change the initially estimated number of social household connections in certain cities, even though demand for connections was less than planned. To avoid such situations, the planned household connections will not be earmarked for particular locations but will follow demand in the project area. It has also been noted that public standposts have been abandoned in favor of house connections since the networks were densified. Thus, the project will provide prepaid public standposts (public water points for which people pay ahead for the service through a vending machine) that can be relocated based on the needs of the area to avoid locking communities in as the network evolves.

27. **The anchoring project management and implementation unit (PMIA) within SOMAPEP-SA guarantees sustainability and ownership, improves performance, and enables building on lessons learned.** The project

²¹ Niger Basin Water Resources Development and Sustainable Ecosystems Management Project (P093806)– Phase1 (2007 – 2017)

²² Through the Sahel Irrigation Initiative Support Project (P154482).

²³ Reference to support of the water and sanitation program on the involvement of the private sector in management of rural water systems and related monitoring systems to measure performance (STEFI).



management performance of the former project (PEMU) is considered to have been good because of close cooperation between SOMAPEP-SA, SOMAGEP-SA, and the Regulatory Commission for Energy and Water. The current project will use the same implementation arrangement, which has delivered good performance and ownership.

28. **Innovative solutions can reduce energy consumption.** The energy bill accounts for 34 percent of SOMAGEP-SA's turnover, which significantly affects the viability of the sector. The financing of solar energy in the PEMU project has alleviated the burden of energy bills in the operating costs of the utility and helped ensure the continuity of water service when the electricity company is failing. The proposed project will apply these learnings to maximize ownership and reduce SOMAGEP operating costs.

29. **Service expansion must be combined with network performance improvement to deliver good-quality services.** The increase in production and pressure in the network after the commissioning of the Kabala treatment plant in 2019 contributed to deterioration in the performance of the Bamako water network, which had been under pressure for a long time, falling from 76 percent in 2019 to 62 percent in 2024. To reduce this decline in performance and thus increase revenues, the project will reduce NRW and strengthen the network.

30. **Alternative solutions should be sought when the enabling environment is not conducive to a specific change. Over the past five years, it has been challenging to advocate for tariff adjustments in Mali.** In an effort to enhance the financial viability of the sector, this project will support initiatives aimed at reducing operational costs, NRW, commercial losses, and network inefficiencies. These measures are intended to increase cost recovery and revenue for the utility, ultimately narrowing the financial gap of the sector. The project team will continue policy dialogue with the government to allow for development of alternative solutions to sustain the financial viability of the urban water sector.

III. PROJECT IMPLEMENTATION

A. Institutional and Implementation Arrangements

31. **This project will build on the successful institutional and implementation arrangement of the Kabala program under the leadership of the Ministry of Energy and Water through SOMAPEP-SA.** The project is anchored in SOMAPEP-SA, which has a strong project management team that has handled several investment projects, including the World Bank-financed PEMU (P122826) in the framework of the Kabala investment program, which has many donors. The institutional and implementation arrangements will ensure proper involvement of other entities benefiting from the project, in particular DNH and SOMAGEP-SA. The institutional framework of the project will be based on a project steering committee (PSC); a project management and implementation agency (PMIA [SOMAPEP-SA]); and two executing agencies (SOMAGEP-SA and DNH) responsible for implementation of Subcomponent 1.3 and Component 2, respectively.

32. **A PSC** chaired by the Ministry of Energy and Water will be responsible for overall oversight and strategic direction of the proposed project. The PSC will be composed of representatives of the Ministry of Economy and Finance (Ministère de l'Economie et des Finances-MEF) through the National Directorate of Development Planning (Direction Nationale de Planification et de Développement- DNP), the General Directorate of Public Debt (Direction Générale de la Dette Publique-DGDP) and the Coordination and Monitoring Unit for Projects and Programs financed by Development Partners, the MEE, the Regulatory Commission for Energy and Water (Commission de Régulation de l'Énergie et de l'Eau- CREE),



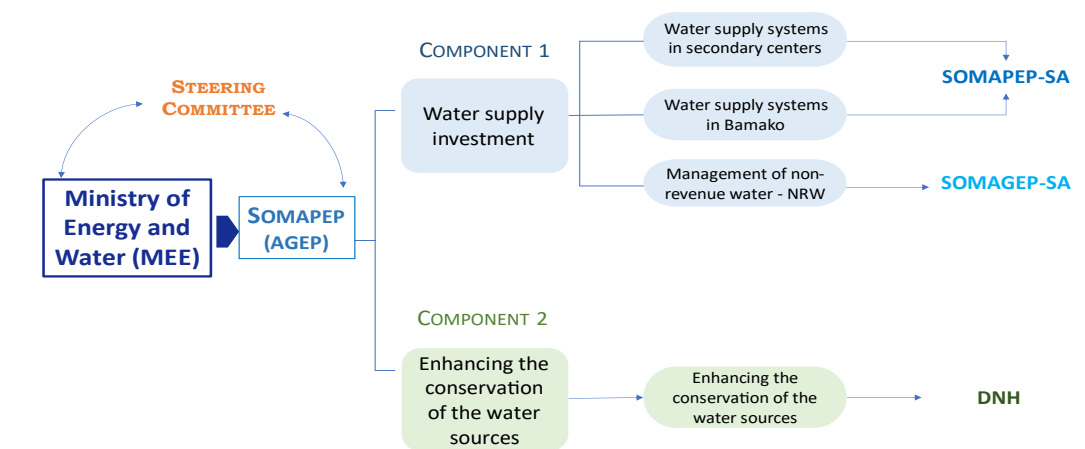
DNH, SOMAPEP-SA, SOMAGEP-SA, and a project coordinator who prepares for PSC meetings and attends them as secretary. The PSC will ensure that activities are consistent with the sector strategy.

33. **SOMAPEP-SA** will be responsible for overall project management; implementation of Subcomponents 1.1 and 1.2; project management and coordination; fiduciary and safeguard activities; M&E; communication activities; preparation of progress and completion reports in accordance with World Bank policies and procedures; payments and disbursements; preparation of annual budgets and work plans; ensuring that annual financial audits are submitted on time; filing and archiving all project documents for auditing purposes; the bidding process for works; supervision of project implementation; coordination of stakeholder activities, including consultants, manufacturers, installers, inspection services, and all public services concerned; approval of technical documents; ensuring sound financial management of the project; coordination of commissioning of infrastructure created by the project once completed; and as secretary, preparation for and participation in PSC meetings. It will also sign a convention with DNH and SOMAPEP-SA to define their roles and responsibilities in project implementation.

34. **SOMAGEP-SA** will manage technical implementation of Subcomponent 1.3 (management of NRW) by hiring, in conjunction with SOMAPEP, consultants to prepare detailed technical studies and bidding documents; helping evaluate bids; and supervising construction related to the subcomponent.

35. **DNH** will manage technical implementation of Component 2 (enhancing conservation of water sources) by hiring, in conjunction with SOMAPEP, consultants to prepare detailed technical studies and bidding documents; helping evaluate bids; and supervising construction related to the component. The institutional and implementation arrangements for the project are depicted in Figure 2 and further described in the relevant sections below.

Figure 2: Institutional Relationships: Mali Water Security Project



B. Results Monitoring, Evaluation, and Verification Arrangements

36. **General characteristics.** The PDO performance indicators and the detailed list of results indicators for the components are provided in Results Framework and Monitoring (Annex 1). This framework consists of expected results, indicators, relevant baseline output and outcome data, milestones, and a suggested timeline for monitoring progress.



The project's monitoring and evaluation (M&E) system has been designed in accordance with this framework and will be available online, with cloud backup, to provide accurate information to verify progress and eventual achievement of results (outputs, outcomes, impacts), support learning during implementation, determine accountability for results, and facilitate informed decision-making.

37. **Technology.** Use of information and communications technologies to facilitate data collection and real-time M&E (use of KoBo Toolbox) will be emphasized. Mobile phones, global positioning systems, piezometers, and tablets will be used to assess management of water resources and water supply systems. The project will incorporate the Geo-Enabling method for Monitoring and Supervision developed by the World Bank Fragility, Conflict, and Violence Team to establish a remote supervision system for operations and increase the capacity of clients to conduct accountable, well-structured M&E. The PMIA will be accountable for integrating Geo-Enabling method for Monitoring and Supervision into the project as part of its responsibility for overall project implementation management, including M&E, safeguards, and E&S standards.

38. **Results measurement for project performance.** A series of baseline studies has been launched as part of preparatory work. The findings of some of these baseline studies have helped refine targets against which project results will be measured. Implementation will be monitored semesterly and annually to facilitate informed strategic decision-making and adjustments as necessary. A mid-term review and an impact evaluation study will be conducted during and at the end of implementation, respectively. The mid-term review will include a qualitative survey to capture women's and men's levels of satisfaction with project investments. The PMIA will be responsible for ensuring strong links between M&E, knowledge management, and strategic communication.

39. **Accountability for results.** In addition to required reporting, the project's M&E system will involve an accountability mechanism comprising stakeholder consultations at the mid-term review. Information-sharing and stakeholder engagement throughout the project cycle will be core components of the project's accountability for results. The PMIA, in coordination with SOMAGEP and DNH, will ensure that stakeholders and beneficiaries have access through various channels to timely, relevant, unambiguous information about the project's M&E findings and are able to bring their views to bear on the project's review and decision-making process.

40. **Harmonization and integration with national and sectoral M&E systems.** The project will make consistent efforts to empower DNH, SOMAGEP, and SOMAPEP to ensure that M&E of project outcomes feeds into national M&E systems (Ministry of Energy and Water).

C. Disbursement Arrangements

41. **Funds will be disbursed under the project according to the Disbursement Guidelines for Investment Project Financing dated May 2017,** the disbursement and financial information letter, and the financing agreement. The designated account (in West African Financial Community francs) will be opened at a commercial bank on terms and conditions acceptable to IDA to facilitate payment for eligible expenditures. The PMIA will manage the designated account according to the disbursement procedures described in the Project Implementation Manual (PIM) and the disbursement letter fiduciary arrangements agreed upon in the subsidiary agreement. IDA will release an initial deposit at the request of the PMIA upon effectiveness. Monthly replenishment will be required, accompanied by the disbursement and procurement plans and relevant supporting documents. The financial management specialist of the PMIA will manage the designated account.

42. **Upon project effectiveness, transaction-based disbursements will be used.** An initial advance up to the ceiling of the designated account will be made into the designated account, and subsequent disbursements will be made monthly



against the submission of statements of expenditures or records as specified in the disbursement letter reporting on the use of the initial or previous advance. Thereafter, the option to disburse against submission of a quarterly unaudited interim financial report (IFR) (also known as report-based disbursements) could be considered subject to the quality and timeliness of the IFRs submitted to the World Bank and the overall financial management performance and arrangements.

IV. PROJECT APPRAISAL SUMMARY

A. Technical, Economic and Financial Analysis (if applicable)

43. **Technical Analysis.** As an extension of the Kabala program in Bamako, the proposed project will use technologies for water supply systems that are well established and have been used effectively in previous water supply projects. The water investment program that the proposed project supports originates from a priority investment program that SOMAPEP-SA designed with the assistance of SOMAGEP-SA to restore water services in critical areas of Bamako and other targeted secondary centers and to maximize the impact of the facilities built under the earlier phases of the Kabala program and those supported by a previous multi-donor Kabala program. The proposed rehabilitation and expansion of water production facilities, feeder pipes, storage and pumping facilities, water distribution networks, and service connections is designed to increase the quantity of water available to existing customers and the quality of service. The social household connections and standpost programs will be conducted in parallel through expansion of secondary and tertiary networks. The proposed project will help provide sector actors with the tools to create financial viability and increase their capacity to conduct their missions. SOMAPEP-SA and SOMAGEP-SA have demonstrated their capacity to implement these programs, including managing expansion projects in close coordination with identification of connection beneficiaries. The government has conducted technical studies for project Subcomponents 1.1, 1.2, and 1.3 in Bamako, Mopti, San, and Sévaré with support from a donor. The World Bank team reviewed these reports, which would expedite subsequent stages and accelerate implementation during the first year of the project. Because the investments under Component 2 will be defined in the first year of project implementation, it is expected that the implementation phase of these investments will start in the second and third years of the project.

Paris Alignment

44. The operation is aligned with the goals of the Paris Agreement on mitigation and adaptation. **The project is consistent with Mali's Nationally Determined Contribution (NDC) as the NDC notes the importance of a low-emission and climate resilient pathway, while emphasizing the importance of providing appropriate solutions to water-related problems.** One key adaptation action included in the NDC is work toward universal access to potable water through creation of new potable water systems, to which this project will directly contribute. Another key component of the NDC is the use of renewable energy, which the project will help tackle through the use of solar powered water pumps. The project is also aligned with the World Bank's G5 Sahel Country Climate and Development Report (CCDR), as the CCDR recommends developing improved integrated water resources management (IWRM) governance and diversifying water sources by leveraging surface and groundwater, both activities included in the proposed project.

45. **Assessment and reduction of mitigation risks.** Activities financed under the project consist of improvements in water supply service provision with an emphasis on solar-powered and energy-efficient solutions, reduction of NRW, and NBSs for water source protection and are therefore universally aligned. A reduction in GHG emissions will be derived from water supply and water source conservation interventions. Mitigation-related activities include reducing NRW by increasing energy efficiency (thus reducing GHG emissions); expanding use of renewable energy, especially solar power through solar-powered groundwater and water conveyance pumping; and NBSs such as reforestation and tree planting that can help sequester carbon. The project will prioritize solar energy for pumping and treatment



whenever economically and technically feasible. If energy demand outstrips solar capacity, grid energy will be prioritized, and diesel generators will be used as a last-case solution.

46. Assessment and reduction of adaptation risks. The main climate and disaster risks affecting the project are drought, flood, and high temperatures. Floods and extreme heat can damage water infrastructure, and droughts can limit the viability of water sources. The project will build flood and drought resilience for users by providing stable, safe drinking water supply systems that are themselves resilient to the impacts of climate change. Climate-resilient measures to be used to protect project investments include placing key infrastructure outside flood zones and areas prone to landslides and using materials that can withstand extreme temperatures. Drought resilience will be ensured by using diverse water sources, including surface water and groundwater. Investments in water infrastructure such as tanks and water distribution networks will consider recommendations from the Resilient Water Infrastructure Design Brief to ensure design that is resilient to drought and flood.

Economic and financial analysis

47. Project benefits. World Bank support will generate significant economic and financial benefits that will enhance the quality of the drinking water service. This will have immediate effects on service continuity while guaranteeing sufficient water for more than 500,000 subscribers. The intervention will generate several other benefits, the most important of which are reduction in water rationing, additional water volume sales resulting from greater water availability thanks to reduction in technical losses, consumer surpluses due to the alternative reduction in the price of water due to network extension and social household connection, reduction in water fetching time for women and girls, and operations and maintenance savings. In addition, improvements in conservation of water sources will involve 30,000 beneficiaries in reforestation and riverbank stabilization and restoration activities.

48. Rationale for public sector financing. In the most recent Country Opinion Survey in Mali (conducted in June/July 2023), stakeholders surveyed identified water supply and sanitation infrastructure a high priority for World Bank Group support to Mali. The public sector's contribution to ensuring development of the drinking water sector is crucial for several reasons: rapid urbanization is exacerbating the existing structural deficit observed in the supply of drinking water services, which justifies continued public financial support for the sector's development; SOMAPEP-SA's revenues are insufficient to sustain a viable self-financing policy of water infrastructure; private sector involvement in financing drinking water infrastructure is limited, as the involvement of several traditional donors in the Kabala program (including World Bank, European Investment Bank, French Development Agency, African Development Bank) underscores; and development of the informal sector is unregulated, leading to a resurgence in water security problems. In light of the above-mentioned development challenges, a strong commitment from and involvement of the public sector and related agencies can increase access to water infrastructure and services. This project is designed to build on previous and ongoing interventions and help the government provide sustainable water services and thereby improve living conditions.

49. Added value of World Bank support. The significant increase in capacity, technical expertise, coordination and use of global knowledge required to provide adequate water infrastructure and services in Bamako and secondary centers justifies the World Bank's support. In addition to building on key achievements and lessons learned from previous and ongoing projects, the World Bank's added value will include direct technical expertise and implementation support from a multidisciplinary team combining expertise from various relevant sectors. Given Mali's significant water infrastructure deficit and the challenge of achieving the Sustainable Development Goals, World Bank support is a key element of its strategic and programmatic approach to supporting sustainable development.



50. **Results of the economic and financial analysis.** The project's economic internal rate of return is 11.3 percent, with a net present value (NPV) at 6 percent discount rate of US\$44.9 million. These indicators show that the economic results that the project will generate are satisfactory. These indicators are robust despite the sensitivity of the economic internal rate of return to a sharp change in investment costs. A 20 percent increase in investment costs lowers the economic internal rate of return to 8.9 percent, and an analysis using financial prices shows that the financial rate of return is significant (15.7 percent), because of the vigorous action taken to manage NRW such as NRW pilot supported by Dutch Cooperation.

Table 2. Economic Analysis Indicators for Mali Water Security Project

Economic internal rate of return	11.3%
NPV @6%	US\$44.9 million
NPV @10%	US\$6.9 million
B/C	1.5

51. **Assessment of impacts of project activities on GHG emissions.** It is estimated that the project will reduce GHG net emissions by 35,400 tons of carbon dioxide equivalent per year (0.928 million tons of carbon dioxide equivalent during the economic lifetime of the project). When shadow prices of carbon (lower and upper limits) are considered to assess the economic viability of the project, the economic internal rate of return of the project rises from 11.3 percent to between 14.5 and 17.8 percent, and the NPV rises from US\$44.9 million to between US\$70.3 million and US\$98.3, which indicates that the project is not only economically viable, but also contributes to the global public good.

Table 3: Project Cost-Benefit Analysis with Economic Impact of Greenhouse Gas Emissions

Low estimate	
NPV @6%	US\$70.3 million
NPV @10%	US\$22.8 million
Internal rate of return	14.5%
High estimate	
NPV @6%	US\$98.3 million
NPV @10%	US\$39.3 million
Internal rate of return	17.8%

B. Fiduciary

(i) Financial Management

52. **The proposed project will be implemented according to applicable World Bank directives, related guidance, and reference materials on investment project financing.** The PMIA inside SOMAPEP-SA, which will have overall financial management responsibility for the proposed project, implemented the closed project PEMU and is implementing the ongoing PRUBA project with Satisfactory financial management performance. The financial management arrangements will be the same as for the ongoing PRUBA project implemented by SOMAPEP-SA. The financial management capacity of the PMIA was assessed in August 2024 according to the financial management manual issued on March 1, 2010, and last revised in September 2021. A memorandum of understanding that clarifies the operational



interactions between the two agencies and PMIA will be signed between the PMIA and each of the partner agencies (DNH, SOMAGEP-SA). This memorandum of understanding will detail the planning, accounting, disbursement, and auditing procedures.

53. Existing financial management arrangements are adequate to provide, with reasonable assurance, accurate and timely financial management information on the project's status that the World Bank requires, although to maintain an adequate financial management arrangement and clear segregation of duties, the existing fiduciary procedures included in the PIM will be updated, and the terms of reference for the key financial management staff (financial management specialist, accountant, internal auditor) will be extended to include the new project before effectiveness. The detailed financial management action plan is presented in Annex 2. The overall financial management risk for the proposed project is Substantial. The detailed risk assessment and the action plan to mitigate the identified risks are presented in Annex 2.

(ii) Procurement

54. **Applicable procurement rules and procedures.** Procurement under the proposed project will be conducted in accordance with the World Bank Procurement Regulations for Investment Project Financing Borrowers, dated November 2016 and revised in September 2023 under the New Procurement Framework; Guidelines on Preventing and Combating Fraud and Corruption in Projects Financed by International Bank for Reconstruction and Development Loans and International Development Agency Credits and Grants, dated October 15, 2006, and revised in January 2011 and on July 1, 2016; and other provisions stipulated in the financial agreements. All procuring entities, bidders, and service providers are required to adhere to the highest ethical standards in awarding and performing contracts financed under the project, in accordance with Paragraph 3.33 and Annex IV of the Public Procurement Regulations.

55. **SOMAPEP-SA, which is staffed with dedicated procurement staff supported by recruited technical experts, and the World Bank team will implement the project.** SOMAPEP-SA will be responsible for day-to-day project management. The project coordinator, as part of the PMIA, holds decision-making authority during the procurement process. Although SOMAPEP-SA has full fiduciary responsibility, SOMAGEP-SA and DNH will be involved in contract management, each within its area of competence. If need be, the recipient can request hands-on expanded implementation support for procurement during project implementation.

56. **Several key elements for procurement must be implemented.**

- Procurement plans and their updates and requests for prior reviews will be sent to the World Bank for clearance through the Systematic Tracking and Evaluation Process (STEP), a planning and tracking system, in accordance with clause 5.9 of the procurement regulations. Procurement activities not requiring World Bank prior reviews will be recorded in STEP as well as the process progresses.
- Procurements that use standard procurement documents will adopt World Bank provisions related to environmental, social (e.g., sexual exploitation and abuse, gender-based violence), and health and safety risks and impacts. This includes codes of conduct with prohibitions against sexual harassment and sexual abuse
- As part of project preparation, the recipient (with World Bank assistance) has prepared a project procurement strategy for development and a detailed procurement plan for the first 18 months acceptable to the Bank. The project procurement strategy for development (PPSD) describes how fit-for-purpose procurement activities will support project operations to achieve development objectives and value for money. The detailed procurement



plan specifies a brief description of the activities and contracts, the selection methods to be used, estimated costs, schedules, World Bank review requirements, and any other relevant procurement information for each contract.

57. Procurement capacity assessments. An assessment of the project implementing entities within the ministries reveals that the main identified risks are related to the country's procurement system, which can lead to delays, inadequate communication, insufficient competition, and poor contract management. The overall procurement risk is Substantial, but it will be regularly assessed and adjusted as needed based on the outcomes of the risk-mitigation measures: training of all project staff involved on procurement regulations, development of a section on procurement procedures as part of the PIM to clarify the roles of team members involved in the procurement process and to define the maximum time frame for each procurement stage, development of contract management plans for major contracts, transfer of key risks (identified in the Procurement Risk Assessment and Management System exercise) to a daily monitoring matrix and monitoring through monthly project implementation meetings with the client during the first two years, and improvement in the filing system to ensure compliance with the World Bank Procurement Filing Manual.

C. Environmental, Social and Legal Operational Policies

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	Yes
Projects in Disputed Area OP 7.60	No

58. The Policy OP/BP 7.50 applies because proposed investments will take place on the Niger River system and the Senegal River system which are international waterways. In accordance with the Policy, other riparian countries were notified of the proposed Project on August 16, 2024, by the Government of Mali through the Niger Basin Authority (NBA) and the Organization for the Development of the Senegal River (OMVS). Both basin organizations confirmed their no objections to the Project and Regional Vice President gave his approval to finalize Project preparation on October 16th, 2024.

59. E&S risk is rated Substantial, and sexual exploitation and abuse and sexual harassment risks are assessed as Moderate. Project activities and their implementation in densely populated urban areas such as Bamako and other secondary cities largely influence the overall risk assessment. Potential risks and impacts of the project during construction may include an increase in pollution from hazardous and non-hazardous waste (e.g., oil, paint), dust, noise, vibrations, and accidental fuel and lubricant spills; traffic congestion that could disrupt local transportation and activities; disturbances of surrounding ecosystems affecting local wildlife and plant life; and health and safety problems for workers and nearby communities, particularly due to working conditions and accident risks. In contrast, during the operational phase, risks and impacts are expected to be minor because management and monitoring measures are appropriate.

60. Given that project sites are clearly defined, it will not be necessary to prepare an E&S management framework to ensure effective management of E&S risks and impacts. Existing E&S impact studies (impact assessments, management plans), conducted for the Mopti/Sévaré and San sites, have been updated to ensure alignment with the



Bank E&S framework. The two updated E&S impact assessments and resettlement action plans were disclosed in country on October 14 and 10, 2024, respectively, and the WB website on October 15 and 14, 2024 respectively. For the sites in Bamako and Kati, specifically the Djicoroni-Para-Missira and Missira-Kati sections, a single E&S impact assessment and management plan will be developed as soon as the necessary technical details are available during project implementation. Given that the technical studies are in progress, this assessment and plan will not be finalized before board approval but will be complete and disclosed once finished. Depending on the conclusions of the E&S impact assessments on the need for involuntary resettlement, the relevant instrument will be prepared. The project will also finance technical assistance to SOMAPEP-SA and SOMAGEP-SA to enable the sector to achieve financial viability and improve technical and commercial performance, including customer management over the entire urban transferred area. These activities will be managed in accordance with the World Bank E&S framework and relevant advisory note for technical assistance. In addition to site-specific instruments, the borrower has prepared labor management procedures and a stakeholder engagement plan. The labor management procedures establish clear protocols for recruitment, training, and management of local workers, ensuring safe and fair working conditions, and address essential aspects such as workplace safety, workers' rights, and conflict resolution mechanisms. The stakeholder engagement plan identifies key stakeholders and addresses their communication needs. This plan supports continuous dialogue with stakeholders, enabling them to voice their concerns and inquire about the project. In consultation with the World Bank, the Borrower has prepared an E&S commitment plan (ESCP) that includes a timeline and a commitment to relevant instruments and sets out the actions that the borrower will perform to meet E&S, health, and safety requirements.

61. Gender. There are significant gender gaps in terms of efforts to gain access to water services and decision-making power in households and communities. Women, especially in rural areas, are responsible for water collection and spend up to eight hours per day on household chores, including up to one hour collecting water. This reduces their time for income-generating activities and education, perpetuating poverty and gender inequality. Women also have limited decision-making power over design and location of water infrastructure, which often exacerbates their disproportionate burden in water collection and increases safety risks, including exposure to gender-based violence when traveling to distant or unsafe water points. To address these challenges, the project will implement two key actions. First, women's input will be actively incorporated into planning and design of water infrastructure, particularly in selection of locations for household connections and standposts. This will be achieved through structured consultations with women's groups, ensuring their participation in decisions regarding water infrastructure placement to enhance accessibility and safety. Designing infrastructure with women's input will help meet their needs, reducing the distance they must travel and enhancing their security. Second, the project will prioritize construction of water points closer to homes, particularly in areas where women are typically responsible for water collection. By decreasing the distance to water sources, the project will reduce the time women spend collecting water, freeing up their time for income-generating activities, education, or other productive uses and increasing their safety by minimizing the risks associated with traveling to distant water points. To monitor progress, the following indicators will be used: percentage of water infrastructure (e.g., household connections and standposts) designed in consultation with women and reduction in average time women spend in water collection, with a target of 30 percent for both. These actions are expected to increase women's participation in water management, reduce their labor burden, and increase gender equality.

62. Citizen engagement. Engaging citizens in project-related activities will be a crucial part of the project. Regular inclusive consultations and participatory planning will be conducted throughout the project cycle. Input from citizens will be integrated into project design and implementation to ensure a closed feedback loop. The project will also ensure that social mobilization and participatory planning are inclusive so that the voices of disadvantaged and vulnerable groups are heard, and all social groups are involved. A gender-related indicator (percentage of water points [household connections and standpoints] designed in consultation with women) will be used to ensure citizen inputs



inform water point design. The project will also develop an accessible, robust GRM to allow beneficiaries and other stakeholders to submit project-related complaints and share concerns and feedback. The GRM will ensure that queries about or clarifications of the project are responded to in a timely manner and that grievances are addressed efficiently and effectively. This will be tracked through the following indicator: grievances addressed in project approved timeline (percentage).

Grievance Redress Services:

63. A project grievance redress mechanism (GRM) will be designed to allow affected stakeholders to submit grievances and seek redress when they perceive that harm has arisen from project interventions. The GRM will be designed in consultation with relevant government and nongovernment stakeholders and will establish accessible processes to submit complaints, as well as clear procedures from investigation to resolution and feedback. The GRM will include a provision for appeal if aggrieved parties are dissatisfied with the outcome. A communication campaign will be implemented to raise awareness and inform stakeholders on how to use the GRM and explain the investigation and sequential resolution process, timeline, and procedures. Communities and individuals who believe that a project that the World Bank supports has harmed them may submit complaints to existing project-level grievance mechanisms or the Bank's Grievance Redress Service, which ensures that complaints received are reviewed promptly to address project-related concerns. Project-affected communities and individuals may submit complaints to the Bank's independent accountability mechanism, which has an inspection panel that determines whether harm occurred, or could occur, as a result of Bank noncompliance with its policies and procedures and a dispute resolution service that gives communities and borrowers the opportunity to address complaints through dispute resolution. Complaints may be submitted to the accountability mechanism at any time after concerns have been brought directly to the attention of Bank management and management has been given an opportunity to respond. For information on how to submit complaints to the Bank's Grievance Redress Service, visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the Bank's accountability mechanism, visit <https://accountability.worldbank.org>.

V. KEY RISKS

64. **The overall risk of the project is rated Substantial** because the political and governance risks are High, and the macroeconomic, sector strategy and policy, fiduciary, E&S, and stakeholder risks are Substantial. Technical design of project or program, institutional capacity for implementation, and sustainability risks are rated Moderate.

65. **The political and governance risk is rated High.** Since the August 2020 and May 2021 events in Mali, the political landscape has remained unstable. The decentralization process has progressed inconsistently, and the project is designed to mitigate associated risks by encouraging implementing agencies (SOMAPEP-SA, SOMAGEP-SA, DNH) to engage in inclusive dialogue with regional and subregional entities and beneficiaries, including vulnerable groups, to ensure smooth implementation of project activities. Addressing political and governance risks through project measures presents significant challenges, which justifies maintaining the residual risk level as High. The unpredictable security situation in the Sahel may divert government attention from development programs, which may affect prioritization of water projects. The project is designed to be flexible, enabling it to adapt to uncertainty and volatility during implementation, and it aligns with the Prevention and Resilience Allocation strategy. A Security Management Plan has been prepared for the project.



66. **The macroeconomic risk is rated Substantial.** The country is facing a challenging macroeconomic environment and outlook because of high security-related spending, supply and demand disruptions, and its exit from the Economic Community of West African States (ECOWAS) in July 2024. The pressures that weak budgetary management and the reduction in agricultural (e.g., in Ségou) and mining production linked to the risk of insecurity exert aggravate macroeconomic instability. The economic outlook is subject to significant downside risks related to the security crisis, the electricity supply crisis, and the withdrawal from ECOWAS. In addition, the tightening of financing conditions in the subregional market increases the cost of financing on the regional market given Mali's high gross financing needs, and the long-standing risk of further climatic shocks remains. These challenges present immediate risks to the project with regard to funding compensation for project-affected people. The government has requested (ref: L/N Cds/fbd/586-24 du 21/11/2024) an exception to be able to fund compensation from IDA as a mitigation measure. A memo was accordingly prepared and approved by the RVP on December 26, 2024)

67. **The sector strategy and policy risk is rated Substantial.** The project is aligned with key sector strategies and policies for water resource management and water supply, but continued delay of the measures that the World Bank has recommended to improve water supply sector financial equilibrium could jeopardize SOMAPEP-SA's and SOMAGEP-SA's ability to take on more projects and realize the objectives of the project. Recent and ongoing dialogue with the Ministry of Energy and Water indicated willingness to implement the tariff study recommendations. The NRW reduction and use of solar energy will contribute to reduce operation cost, increase the sector revenue and therefore improve the financial viability of the utilities.

68. **The fiduciary risk is rated Substantial** despite the experience that SOMAPEP-SA gained in Bank operations with PEMU (Kabala) and the ongoing PRUBA. The project design, with multiple agencies, combined with the current country context, presents some challenges from a fiduciary standpoint. Proposed mitigation measures include the renewal of the previous PEMU project team located within SOMAPEP-SA, which already has a procurement function run by a tried and tested unit with experienced fiduciary and safeguards experts familiar with World Bank procedures (financial management, procurement, internal and external audit, safeguards, M&E), and draft of the PIM, including financial, administrative, accounting, technical, and safeguards procedures.

69. **The E&S risks are rated Substantial.** The activities proposed under Components 1 and 2 may entail risks to the health and safety of local populations in densely populated areas of several districts of Bamako and some secondary towns, risks of land acquisition and loss of private property, and businesses, and risks of disruption of social activities such as movement of pedestrians and cars and parking. Mitigation measures are included in the ESIAs/RAPs prepared for Mopti/Sevare and San.

Project Readiness

70. Given the extensive engineering studies and safeguard analyses already conducted for the Bamako water supply system and several secondary cities, project readiness is advanced. Most of the activities that the project supports are at the detailed design stage, are assessed for half of Bank financing (US\$52 million equivalent), and are ready to be implemented shortly after effectiveness. The main safeguard documents (ESIAs/RAPs of Mopti/Sevare and San) required before work starts have been approved and disclosed. In accordance with Bank Operating Procedure 7.50, which governs projects involving international waterways, riparian countries were duly notified through the appropriate international river basin agencies (Senegal River Development Organization, Niger Basin Authority). Neither agency issued any objections, because the volumes to be abstracted are minimal in relation to the overall water resources, and no significant impact is anticipated on these two international waterways. The RVP was notified, and approval was given on October 10, 2024 to continue with project preparation.



ANNEX 1. RESULTS FRAMEWORK

PDO Indicators by PDO Outcomes

Baseline	Period 1	Closing Period
Increase access to improved drinking water supply in selected cities		
People provided with at least basic water (Number of people) ^{CRI}		
Sep/2024	Dec/2027	Dec/2029
0	180,000	227,000
➤ People provided with at least basic water - Female (Number of people) ^{CRI}		
Sep/2024	Dec/2027	Dec/2029
0	90,000	113,500
➤ People provided with at least basic water - Youth (Number of people) ^{CRI}		
Sep/2024	Dec/2027	Dec/2029
0	34,200	43,130
Operating cost ratio of SOMAGEP improved (Percentage)		
Sep/2024	Dec/2027	Dec/2029
116	105	100
Lowered non-revenue water (NRW) (Percentage)		
Sep/2024	Dec/2027	Dec/2029
40	37	35
Enhance conservation of water sources in selected cities.		
Landscapes under enhanced conservation and/or sustainable management (terrestrial and inland water areas) (Hectare(Ha)) ^{CRI}		
Sep/2024	Dec/2027	Dec/2029
0	1,500	3,000

Intermediate Indicators by Components

Baseline	Period 1	Closing Period
Component 1: Water supply in Bamako and selected secondary centers		
Storage Capacity (Cubic Meter(m3))		



Sep/2024	Dec/2027	Dec/2029
0	6,000	9,000
Household connections constructed under the project (Number)		
Sep/2024	Dec/2027	Dec/2029
0	12,000	20,000
Standpipes built/rehabilitated (Number)		
Sep/2024	Dec/2027	Dec/2029
0	50	90
Leaks detected and treated/monthly on average (Number)		
Sep/2024	Dec/2027	Dec/2029
2,000	1,300	1,000
People with access to improved water service (Number)		
Sep/2024	Dec/2027	Dec/2029
0	170,000	270,000
Grievances addressed in project approved timeline (Percentage)		
Sep/2024	Dec/2027	Dec/2029
76	82	90
Reduction in the average time women spend in water collection (Percentage)		
Sep/2024	Dec/2027	Dec/2029
0	18	30
Component 2: Enhancing the conservation of the water sources		
Monitoring network for water resource quantity and quality established (Yes/No)		
Sep/2024		Dec/2029
No		Yes
River-bank stabilized (Meter(m))		
Sep/2024	Dec/2027	Dec/2029
0	2,000	5,000
Component 3: Project management and Capacity reinforcement		
Component 4: Contingency Emergency Response Component		



Monitoring & Evaluation Plan: PDO Indicators by PDO Outcomes

Increase access to improved drinking water supply in selected cities	
Lowered non-revenue water (NRW) (Percentage)	
Description	This indicator measures the reduction of non-revenue water (NRV) of the water distribution in terms of network efficiency and bill collection rate.
Frequency	Every six months
Data source	SOMAGEP
Methodology for Data Collection	Measurement of volume of water produced and water sales. The indicator is the ratio of water sales to water produced
Responsibility for Data Collection	SOMAGEP
People provided with at least basic water (Number of people) ^{CRI}	
Description	No description available for this Corporate indicator
Frequency	Every six months
Data source	SOMAPEP
Methodology for Data Collection	Based on household connection and public standpost. Beneficiaries are 10 persons for each household connection and 300 persons for each public standpost
Responsibility for Data Collection	SOMAPEP
People provided with at least basic water - Female (Number of people) ^{CRI}	
Description	No description available for this corporate indicator
Frequency	Every six months
Data source	SOMAPEP
Methodology for Data Collection	Based on household connection and public standpost. Beneficiaries are 10 persons for each household connection and 300 persons for each public standpost
Responsibility for Data Collection	SOMAPEP
People provided with at least basic water - Youth (Number of people) ^{CRI}	
Description	No description available for this corporate indicator
Frequency	Every six months
Data source	SOMAPEP
Methodology for Data Collection	Based on household connection and public standpost. Beneficiaries are 10 persons for each household connection and 300 persons for each public standpost
Responsibility for Data Collection	SOMAPEP
Operating cost ratio of SOMAGEP improved (Percentage)	
Description	This indicator measures the operating efficiency of the water utility SOMAGEP. This is measured through the ratio between the operating cost and the turnover. more the ration is under 100% better is the operating efficiency.
Frequency	Every year
Data source	SOMAGEP
Methodology for Data Collection	Ratio of operating costs to revenues.
Responsibility for Data Collection	SOMAGEP
Enhance conservation of water sources in selected cities.	
Landscapes under enhanced conservation and/or sustainable management (terrestrial and inland water areas) (Hectare(Ha)) ^{CRI}	
Description	This indicator measures the drought or flood prone surface area under enhanced conservation and management to protect water sources by recharging the groundwater. It includes areas under soil stabilization and reforestation, and



	under enhanced natural aquifer recharge.
Frequency	Every six months
Data source	DNH
Methodology for Data Collection	Measurement of areas under soil stabilization and reforestation and enhanced natural aquifer recharge because of the project
Responsibility for Data Collection	DNH

Monitoring & Evaluation Plan: Intermediate Results Indicators by Components

Component 1: Water Supply investment	
Storage capacity achieved (Cubic Meter(m3))	
Description	Additional storage capacity attributable to the project: 2,000 m ³ for Bamako, 5,000 m ³ for Mopti-Sévaré, and 2,000 m ³ for San (Gao not considered yet)
Frequency	Every six months
Data source	SOMAPEP
Methodology for Data Collection	Capacity of water tanks built
Responsibility for Data Collection	SOMAPEP
Household connections constructed under the project (Number)	
Description	the indicator measure the number of households benefiting from household connection through social connections financed by the project
Frequency	Every six months
Data source	SOMAPEP
Methodology for Data Collection	Capacity of water tanks built
Responsibility for Data Collection	SOMAPEP
Standpipes built/rehabilitated (Number)	
Description	This indicator measures the number of standpipes built to allow people to access water for a time of round trip that will not exceed 30 minutes and in consultation with beneficiaries.
Frequency	Every six months
Data source	SOMAPEP
Methodology for Data Collection	Number of household connections built under the project
Responsibility for Data Collection	SOMAPEP
number of leaks detected and treated/monthly on average (Number)	
Description	This indicator measures the average number of leaks detected and treated monthly
Frequency	Every six months
Data source	SOMAGEP
Methodology for Data Collection	Leaks treated monthly
Responsibility for Data Collection	SOMAGEP
percentage of complaint received and addressed monthly (Percentage)	
Description	This indicator measures the percentage of complaint received and addressed per month. This indicator is monitored by SOMAGEP according to the utility's reference that aimed to appropriately address complaints in 48 hours following the reception.



Frequency	Every six months
Data source	SOMAGEP
Methodology for Data Collection	Percentage of complaints received per month that are addressed within 48 hours
Responsibility for Data Collection	SOMAGEP
Reduction in the average time women spend in water collection (Percentage)	
Description	This indicator measures the average time women spent for water collection before and after the project. it is anticipated that at least 30% of women save time in their water collection after the project. Mechanisms to measure this could be: •Pre- and post-project surveys: This could entail conducting surveys among women in the targeted communities to measure changes in their access to water, satisfaction with the infrastructure, and perceived safety. These results would be compared to baseline data to assess the impact of women's involvement in the design process. •Focus groups and interviews: Focus groups and interviews with women in the community to gather qualitative data on their experiences with the new water infrastructure. Asking about the impact on their daily lives, safety, and overall well-being. •Women's committees: If at all feasible, if women's committees/ cooperatives at the community level could be established or existing ones strengthened, this would be a very strong approach to enabling women to be involved in all stages of the project, from decision-making and construction to oversight and monitoring. This would also allow for the quantification of participation in the design of water infrastructure.
Frequency	Three times during project life: baseline, mid-term, end project
Data source	SOMAPEP
Methodology for Data Collection	Survey, focus group
Responsibility for Data Collection	SOMAPEP
People with access to improved water services (Number)	
Description	Number of people who had limited access to water services that benefited from improved access (24-hour access with appropriate water pressure) because of the project
Frequency	Every six months
Data source	SOMAGEP
Methodology for Data Collection	Number of people with improved service (daily service duration, water pressure and quantity)
Responsibility for Data Collection	SOMAGEP
Component 2: Enhancing the conservation of the water sources	
Linear length of river-bank stabilized (Meter(m))	
Description	Length of river banks stabilized to safeguard the land and related ecosystems
Frequency	Every six months
Data source	DNH
Methodology for Data Collection	Length of riverbank stabilized.
Responsibility for Data Collection	DNH
Monitoring network for water resource quantity and quality established (Yes/No)	
Description	Establishment and functioning of a monitoring system between SOMAGEP/SOMAPEP and DNH to monitor water



	abstraction and quality
Frequency	Every six months
Data source	DNH
Methodology for Data Collection	Functionality of monitoring network
Responsibility for Data Collection	DNH
Component 3: Project management and Capacity reinforcement	
Component 4: Contingency Emergency Response Component	



ANNEX 2: IMPLEMENTATION ARRANGEMENTS AND SUPPORT PLAN

Financial Management Assessment Report

1. **The PMIAPMIA in SOMAPEP, which will have overall financial management responsibility for the proposed project, implemented the Mali Urban Water Project (PEMU, P122826), which closed in December 2022, and is implementing the ongoing Bamako Urban Resilience Project (P171658), with Satisfactory financial management performance.** The financial management arrangements will be the same as those for the ongoing project. The financial management capacity of the PMIA was assessed in August 2024, according to the Financial Management Practices Manual issued on March 1, 2010, and last revised in September 2021.
2. **The overall financial management risk for the proposed project is Substantial considering the country public financial management context.** At the SOMAPEP level, existing financial management arrangements are adequate to provide, with reasonable assurance, the accurate and timely financial management information on the project's status that the World Bank requires, although to maintain an adequate internal control environment and clear segregation of duties, the negotiation will update the existing fiduciary procedures of the Project Implementation Manual (PIM) of the PMIA to reflect the proposed project's activities. The assessment revealed that the financial management arrangements meet the Bank's minimum financial management requirements under Bank Policy and Directive for Investment Project Financing operations. Subject to implementation of the financial management action plan as outlined below, they are adequate to provide, with reasonable assurance, the accurate and timely information on the project's status that the World Bank requires. The overall financial management residual risk rating is Substantial.

Risk assessment and mitigation

3. The World Bank financial management team assessed the financial management of the implementing entity SOMAPEP in August 2024. The World Bank's principal concern is to ensure that project funds are used economically and efficiently for the intended purposes. Assessing the risks of the project funds not being used is an important part of the financial management assessment work.
4. The assessment revealed that SOMAPEP has sufficient financial management capacity to meet World Bank requirements. The overall financial management residual risk for the project is rated Substantial, considering project design and country context. The PMIA could be in a position to manage the World Bank's funds once the following measures are implemented before and after project effectiveness. The content of these risks is described in the table below.
5. The overall fiduciary risk is rated Substantial, and mitigation measures proposed (see financial management Action Plan) will strengthen the internal control environment and maintain the continuous timeliness and reliability of information that the PMIA produces and adequate segregation of duties.

Table 2.1: Financial Management Risk Assessment and Mitigation

Risk	Inherent risk	Risk-mitigating measures incorporated into project design	Residual risk
Country level: Public Expenditure and Financial Accountability II, undertaken in 2021, has highlighted several areas of strength.	High	The government is committed to a reform program to improve public financial management, and the Modernization of Public Financial Management and budget program were established in January 2018, although it will take time for these reforms to substantially improve public fund management.	Substantial
Entity level: Lack of capacity for implementing	Moderate	The Malian Potable Water Heritage Company is familiar with IDA financial management procedures. The financial	Moderate



Risk	Inherent risk	Risk-mitigating measures incorporated into project design	Residual risk
entity can delay project implementation.		management procedures manual will be prepared. Financial management staff familiar with the World Bank will be retained and strengthened if needed. Accounting software will be installed.	
Project level: Project resources might not be used for intended purposes. Delays in reporting system and auditing due to the additional workload for and the limited capacity of the financial management team are expected. The large number of stakeholders could hinder implementation of the project. Other concerns are the weakness of the monitoring and evaluation system to support payment of cash grants and limited capacity of regional and rural entities.	Substantial	Current financial management arrangements under the Bamako Urban Resilience Project are adequate to manage the project. For efficiency, a financial specialist will strengthen ex-ante and ex-post control of funds allocated to partner implementing organizations. The scope of audit will include review of expenditures that implementing entities incur. Specific measures are incorporated into project design to ensure smooth implementation and mitigate related risks, including governance actions.	Substantial
Control risk	Substantial		Substantial
Budgeting: Limited capacity to prepare and submit accurate work program and budget, budgets not consolidated, weak budgetary execution and control, delays in preparing budget, unreliable lack of comprehensiveness of budget, cost overruns or underruns and reasons not detected in timely manner	Substantial	Annual work plan and budget is required each year and proclaimed, and the project steering committee reviews and approves it. The Project Financial Procedures Manual will define the arrangements for budgeting, budgetary control, and requirements for budgeting revisions. An interim financial report will provide information on budgetary execution and analysis of variances between actual and budget.	Moderate
Accounting: Poor policies and procedures, lack of qualified accounting staff (staff capacity); and lack of familiarity with the accounting system for the Organization for the Harmonization of Business Law in Africa.	Substantial	Financial management aspects handled by the financial management team of the PMIA. The project will adopt the accounting system for the Organization for the Harmonization of Business Law in Africa. Accounting procedures will be documented in the procedures manual.) The financial management team, headed by an administrative and financial specialist, will be strengthened by individual consultants recruited competitively. Training on IDA financial management procedures will be provided to staff as needed.	Moderate
Internal Control: Internal control system may be weak because of lack of financial management capacity of project team, or current financial management procedures may not be adequate for this project. Lack of a procedural	Substantial	Prepare financial management procedures manual and training on use of manual. Outsource internal audit function to a consultant who will scrutinize the proclaimed accounting, financial, and operational procedures. Internal auditor will report to the project coordinator and share the report with the Project Steering Committee.	Moderate



Risk	Inherent risk	Risk-mitigating measures incorporated into project design	Residual risk
manual may lead to inappropriate use of funds and delays in financial reporting. The Project Steering Committee may not be effective.			
Funds Flow: Risk of misuse of funds and delays in disbursements of funds to internal auditor and beneficiaries; inefficient use of funds; delay in use of advances; delay in justification of use of advance made to internal auditor	Substantial	In line with financial management manual, payment requests are to be prepared before disbursement of funds to contractors or consultants and implementing entities. (Terms of reference of internal and external auditors include regular field visits (physical verifications of projects, goods, and services acquired). Ceiling for expenditures that can be handled or paid in cash will be set up in the financial management procedures manual Bank accounts will be replenished via a simplified IFR (summary report). Supporting documents will be kept on their premises.	Moderate
Financial reporting: Inaccurate and delayed submission of IFR; workload leading to some delays in recording of expenditures and preparation of periodic financial reports Lack of motivation of staff working on IDA-financed project affecting internal control and quality of oversight of IDA funds by implementing entities	Substantial	Computerized accounting system will be used (e.g., multi-projects and multi-sites) IFR and financial statement formats were agreed upon at Project negotiations. One finance manager will lead PMIA financial management team PMIA, and one additional accountant will be recruited for project.	Moderate
Auditing: Delays in submission of audit report Mission's scope may not cover expenditures incurred at decentralized level and other internal auditors, and the auditors selected may not be acceptable to IDA or might not conduct their assignments professionally.	Substantial	Project's institutional arrangements allow appointment of adequate external auditors. Terms of reference (to be discussed before expression of interest advertised) will include field visits and specific reports on finding physical controls of goods, services, and works acquired by internal auditors and beneficiaries Annual auditing arrangements will be conducted during project implementation period World Bank financial management team will closely monitor audit due dates.	Moderate
Fraud and corruption: Possibility of circumventing internal control system with colluding practices such as bribes and abuse of administrative positions. Mis-procurement is a critical issue.	Substantial	Terms of reference of external auditor will comprise a specific chapter on corruption auditing Internal auditor will report to Project Steering Committee Copies of the internal auditor reports will be submitted to World Bank Measures to increase transparency, such as providing information on project status to the public, and to encourage participation of civil society, beneficiaries, and other stakeholders are built into project design (see section below on Guidelines on Anti-Corruption).	Substantial
OVERALL financial management RISK			Substantial

Note: IDA, International Development Agency; IFR, interim financial report.



Financial management action plan

6. The action plan outlined in Table 2.2 below indicates the actions that must be taken to strengthen the project's financial management system.

Table 2.2. Financial Management Action Plan

Issue or topic	Remedial action recommended	Responsible body	Completion date	Financial management effectiveness conditions
Staffing	Extend current Bamako Urban Resilience Project financial management staff (RAF and accountant) contract to cover the proposed Project	SOMAPEP	Before negotiation	No
	Update internal auditor terms of reference and working program to include the project	SOMAPEP		
	Recruit an accounting assistant	SOMAPEP	3 months after effectiveness	
Information system accounting software	Update accounting software	SOMAPEP		
Internal control	Update Project Implementation Manual, including financial, administrative, and accounting procedures	SOMAPEP	Before negotiation	
External auditing	Appoint external auditor and sign contract	SOMAPEP	6 months after effectiveness	

Note: SOMAPEP, Malian Assets Holding Company (*Société Malienne de Patrimoine de l'Eau Potable*).

Planning and Budgeting Arrangements

7. The planning and budgeting processes of the project will follow the guidance detailed in Instruction No. 14-2628/MEF-SG, signed by the Ministry of Economy and Finance on August 12, 2014. For this project, the PMIA will coordinate preparation of a consolidated annual work plan and budget to implement project activities, considering the project's components. The work plan and budget will identify the activities to be undertaken, and the roles of the respective parties involved in the implementation, including all the implementing entities, and will provide detailed information on the amount allocated to each implementing entity per activity, showing unit costs and quantities. The PMIA will consolidate annual work plans and budgets into a single document and submit it to the Project Technical Committee and thereafter to the World Bank for approval. There should be no objection later than November 30 of the year preceding the year that the work plan should be implemented.

Key Accounting Policies and Accounting Software



8. The financial management unit of the PMIA will implement and use an integrated financial and accounting system PMIA. The project code and chart of accounts will be developed to meet the project's needs and documented in the manual of procedures being drafted. The prevailing accounting policies and procedures in line with the West African Francophone countries' accounting standards—the Organization for the Harmonization of Business Law in Africa—currently in use in ongoing World Bank–financed operations in Mali will apply. The project's administrative, accounting, and financial manual will document the accounting systems, policies, and financial procedures.
9. For the project to deliver its objectives, a computerized financial management system capable of recording transactions and reporting project operations in a timely manner, including preparing withdrawal applications and periodic financial reports (interim financial reports [IFRs], annual financial statements), will be purchased and customized. The system should combine budgeting, operating, and cost accounting systems to facilitate monitoring, evaluation, and reporting.

Internal Control and Auditing

10. The internal control system is intended to ensure effectiveness and efficiency of operations, reliability of financial reporting, and compliance with applicable laws and regulations. For this project, the accounting, financial, and administrative procedures manual to be developed will document, explain, and describe work processes, information flow, authorization and delegation of authority, timing, job segregations, automatic and sequential controls, compliance with project objectives, and micro and macro rules and regulations. Applying the procedures in the manual will be mandatory for all staff at all levels.
11. The internal auditor (individual consultant) at the PMIA will review project expenditures monthly, including physical verification of acquisition of projects, goods, and services. The project coordinator will communicate all deficiencies or circumvented practices identified for immediate corrective action as appropriate. One copy of such report also will be communicated to the World Bank and the steering committee.
12. A memorandum of understanding will be signed between the PMIA and each of the partner agencies (National Directorate of Hydraulics [*Direction Nationale de l'Hydraulique*, DNH], SOMAGEP), based on an evaluation questionnaire to be completed before the agreements are signed, based on the risks identified. This memorandum of understanding will detail the planning, accounting, disbursement, and auditing procedures.

Disbursements and Fund Flows

13. Funds will be disbursed under the project according to the provisions of the Disbursement Guidelines for Investment Project Financing dated May 2017, the disbursement and financial information letter, and the financing agreement.
14. The designated account (in West African Financial Community francs) will be opened at the commercial bank on terms and conditions acceptable to IDA to facilitate payment for eligible expenditures. The PMIA will manage the designated account according to the disbursement procedures described in the PIM and the disbursement letter. Monthly replenishment will be required, accompanied by the disbursement and procurement plans and the relevant supporting documents. The PMIA financial management specialist PMIA will manage the designated account.
15. These accounts will be managed according to the disbursement procedures described in the PIM and fiduciary arrangements included in the subsidiary agreement. IDA will release an initial deposit at the request of the PMIA upon effectiveness. Implementing partner organizations (e.g., firms, contractors, nongovernmental organizations,



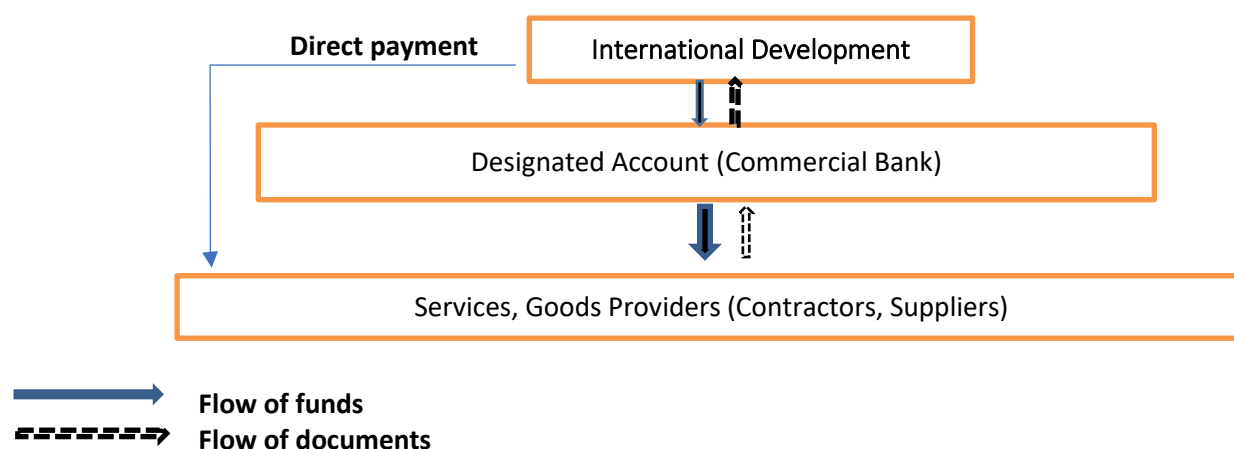
specialized government entities), service providers, and suppliers will be paid in accordance with the payment's modalities, as specified in the respective contracts and the memorandum of understanding. In the case of payments to UN agencies, the special World Bank disbursement procedures will be used to establish a blanket commitment that will be established for the UN agency to transfer the full amount to the UN agency as an advance. The credit will finance all of the project's eligible expenditures, including taxes.

16. Upon project effectiveness, transaction-based disbursements will be used. An initial advance up to the ceiling of the designated account will be deposited into the designated account, and subsequent disbursements will be made monthly against submission of statements of expenditures or records as specified in the disbursement letter reporting on use of the initial or previous advance. Thereafter, the option to disburse against submission of quarterly unaudited IFRs (also known as report-based disbursements) could be considered subject to the quality and timeliness of the IFRs submitted to the Bank and the overall financial management performance and arrangements as assessed in due course. In addition to the advance method, for contracts above a predetermined threshold for eligible expenditures (e.g., 20 percent of the designated account ceiling), the option of disbursing the funds through direct payments to a third party will also be available. Another acceptable method of withdrawing proceeds from the IDA credit is the special commitment method. IDA may pay amounts to a third party for eligible expenditures that the recipient pays under an irrevocable letter of credit.

Flow of Funds

17. Fund flows will be from the designated account to suppliers' and contractors' accounts (Figure 2.2).

Figure 2.2. Flow of Funds



Local Taxes

18. Funds will be disbursed in accordance with project categories of expenditures and components, as shown in the financing agreement. Financing of each category of expenditure or component will be authorized inclusive of all taxes according to the current country financing parameters approved for Mali.

Financial Reports

19. Financial reports will be designed to provide timely, good-quality information on project performance to project management, the World Bank, and other relevant stakeholders. The PMIA will submit quarterly IFRs, which should



include a statement of sources of funds and project revenues and uses of funds; a statement of expenditures classified according to project component or disbursement category (with additional information on expenditure types and implementing agencies as appropriate), showing comparisons with budgets for the reporting quarter, the year, and cumulatively for the project life; a cash forecast; explanatory notes; and designated account activity statements. The quarterly IFR will be prepared and submitted to the World Bank within 45 days after the end of each calendar quarter and will reflect the activities that the PMIA or any UN agency or other implementing entity implements.

20. In compliance with international accounting standards and IDA requirements, the project will produce annual consolidated financial statements under the responsibility of PMIA, including a balance sheet that shows assets and liabilities; a statement of sources and uses of funds showing all the sources of project funds and expenditures analyzed according to project component or category; a designated account activity statement; a summary of withdrawals using statements of expenses listing individual withdrawal applications according to reference number, date, and amount; and notes related to significant accounting policies and accounting standards that management has adopted and that underlie the preparation of financial statements. The financial statements will constitute the entry point of the external auditor's annual diligence.

Auditing Arrangements

21. The supreme audit institution (Section of Accounts of the Supreme Court) has limited capacity (e.g., human and financial resources), so a qualified external audit firm will be recruited under terms of reference and procedures acceptable to the World Bank and will audit the project's financial statements annually. The PMIA will prepare the terms of reference for the audit, the selection process will be launched, and the external auditor will be recruited not later than six months after project effectiveness. The audit will cover the activities that the implementation agencies and any partner implementing agencies implement at the central and decentralized levels.
22. The annual audited financial statements, together with the auditor's report and management letter covering identified internal control weaknesses, will be submitted to the World Bank no later than six months after the end of each fiscal year. A single audit opinion will be issued that covers project income and expenditures, designated accounts, and the IFR. The report will also include specific controls such as compliance with procurement procedures and financial reporting requirements and consistency of financial statements, management reports, and field visits (e.g., physical control). The audit report will thus refer to any incidence of noncompliance, ineligible expenditures, or misuse identified during the audit mission.

Table 2.3: Audit Reports

Audit report	Due date
Project audit reports (audit report and management letter)	Not later than June 30 (Year N) if effectiveness occurred before June 30 (Year N-1). Not later than June 30 (N+1) if effectiveness has occurred after June 30, (N-1)

Implementation Support Plan

23. Based on the outcome of the financial management risk assessment, an implementation support plan is proposed (Table 2.4), the objective of which is to ensure that the project maintains a satisfactory financial management system throughout its life.



Table 2.4: Financial Management Implementation Support Plan

Financial management activity	Frequency
Desk reviews	
Interim financial report review	Quarterly
Audit report review of program	Annually
Review of relevant information such as interim internal control systems	Continuous as available
On-site visits	
Review of overall operation of the financial management system (support missions)	Twice per year
Monitoring of actions taken on issues highlighted in audit reports	As needed
Transaction reviews	As needed
Capacity-building support	
Financial management training sessions	During implementation, as needed

Governance Anticorruption Measures

24. The risk of irregularities and corruption within project activities is substantial given the country context and health sector performance. Lack of appropriate or effective oversight mechanisms could also jeopardize project implementation. A strong fiduciary arrangement has been designed and established to mitigate these risks. Some measures to increase transparency include publishing project audited financial statements online and recruiting a dedicated financial management officer and an internal auditor familiar with World Bank financial management procedures.

PROCUREMENT

25. **The recipient will conduct procurement for the proposed project in accordance with World Bank Procurement Regulations for Investment Project Financing Borrowers** September 2023 under the New Procurement Framework; the Guidelines on Preventing and Combating Fraud and Corruption in Projects Financed by the International Bank for Reconstruction and Development Loans and International Development Association Credits and Grants, dated October 15, 2006, and revised in January 2011 and on July 1, 2016; and other provisions stipulated in the financing agreements. All procuring entities, bidders, and service providers (suppliers, contractors, consultants) shall observe the highest standard of ethics during procurement and execution of contracts financed under the project in accordance with Paragraph 3.32 and Annex IV of the procurement regulations.
26. **Systematic Tracking of Exchanges in Procurement.** The proposed project shall use Systematic Tracking of Exchanges in Procurement (STEP) as a procurement planning and tracking tool to prepare, clear, and update its procurement plans and conduct all procurement transactions between the implementing agency and the World Bank. Use of STEP is mandatory for all procurement transactions subject to the Bank's post and prior review under the project. The contracts will thus be recorded in and processed through STEP. This arrangement will provide data on procurement activities, establish benchmarks, monitor delays, and measure procurement performance.
27. **Project procurement strategy for development:** As part of preparation, the recipient has prepared a project procurement strategy for development (PPSD) that is acceptable to the Bank. This PPCSD describes how fit-for-purpose procurement activities will support project operations for achievement of project development objectives and deliver value for money. The procurement strategy is linked to the project implementation strategy at the regional, country, and state level, ensuring proper sequencing of activities. It considers institutional arrangements for procurement and includes a detailed assessment and description of state government capacity for conducting procurement and



managing contract implementation within an acceptable governance structure and accountability framework. Other things considered include behaviors, trends, and capabilities of the market (market analysis) to inform the procurement plan.

28. Special arrangements such as direct contracting, use of statements of expenditures, third-party monitors, local nongovernmental organizations, force accounts, civil servants' needs, results-based arrangements, need for prequalification, and public-private partnership contracts, if any, will be considered and addressed. The PPSD also describes the selection methods to be followed in the procurement of goods, projects, and non-consulting and consulting services financed under the project.
29. Using the PPSD conclusion, the recipient prepared a detailed 18-month procurement plan, which specifies, for each contract, a brief description of the activities and contracts, the selection methods to be used, the estimated cost, schedules, World Bank review requirements, and any other relevant procurement information. Through the course of project implementation, the procurement plan will be updated as required (at least annually) to reflect project implementation needs.
30. The recipient is required to prepare and submit a general procurement notice (GPN) to the World Bank, which will arrange for its publication in the United Nations Development Business online platform and on its external website. The recipient is also encouraged to publish the general procurement notice in at least one national newspaper with a large circulation, ensuring maximum visibility.
31. The PMIA plays a crucial role in the project's planning and procurement management and is responsible for day-to-day management of the project. The project coordinator, as part of the PMIA, holds authority for decision-making during the procurement process. The PMIA will regularly update the PIM, including the section on procurement procedures. This manual will provide a detailed description of the processes and procedures related to implementation of the proposed activities and set out detailed procedures for maintaining and providing access to project procurement records, in compliance with the financing agreement.
32. Procurement capacity and risk is assessed, mainly through the PPSD analyses which enables public contracts to be properly executed because reforms were implemented to improve management of public contracts. The various texts adopted since 2014 have maintained the elimination of duplication of prior review; extended the scope of the code to all contracts, depending on the amount; and offer new procurement methods. In addition, the World Bank has assessed the country's regulations and considers that the country's new procurement procedures comply with Bank guidelines.
33. However, challenges remain, and experience has shown that most dedicated public officials have little mastery of the rules and procedures governing award and management of public contracts and are exposed to hierarchical influences in decisions; significant delays are recorded in the procurement process, especially in preparation of tender documents; most of the delays are in the contract approval and signature stage within the government; and technical evaluation reports of proposals tend to be of poor quality. The overall procurement risk is Substantial, but after implementation of the proposed mitigation measures, including on adequate staffing, the risk will be Moderate.
34. An in-depth assessment was conducted during preparation of the PPSD. In consultation with the borrower, an action plan to mitigate the risks identified in the procurement assessment has been prepared (Table 2.5). With implementation of the measures proposed in the action plan and support from the World Bank team, the overall procurement risk could be mitigated to Moderate.



Table 2.5: Action Plan for Strengthening Procurement Capacity (to Be Completed after Finalization of Project Procurement Strategy for Development)

No.	Key risks	Mitigation actions	Responsible entity	Timeline
1.	Lack of mastery of World Bank New Procurement Framework and its tools by procurement team and technical staff involved in procurement processes	Recruit procurement team (procurement specialist, procurement assistant), technical experts, Directorate of Finances and Materials staff, and tender committee members and train them on World Bank New Procurement Framework.	PMIA	Within 3 months after credit effectiveness
2	Timeouts in implementation of some activities, mainly evaluation committee management and awarding of contracts	Closely monitor quality and exercise control of all aspects of procurement process, including evaluation, selection, and contract award, in line with provisions of procurement manual.	PMIA	Throughout project implementation
3	Delay in delivery time, which could affect activity schedule	Systematically provide a dissuasive application clause for late payment penalties and apply it if necessary. Ensure technical and financial capacity of supplier.	PMIA	Throughout project implementation
4	Poor description of supplies during preparation of tender file	Involve competent public service or experts in the field in preparation of file. Involve International Development Agency in quality review of technical specifications.	PMIA	Throughout project implementation
5	Delay in publishing tender documents and processing procurement files	Involve Coordination, Project and Program Monitoring Unit in obtaining legal opinion of General Directorate of Public Procurement and Public Service Delegations.	PMIA	Throughout project implementation

Note: Project Implementation unit is the entity responsible for these mitigation actions.

35. Procurement methods. The recipient will use procurement methods and market approach in accordance with the procurement regulations. Procurement will therefore follow direct selection; use of framework agreements, including existing ones; procurement from UN agencies; and increased thresholds for requests for quotations and national procurement. The open national market approach is a competitive bidding procedure normally used for public procurement in the country of the recipient and may be used to procure goods, projects, or non-consultant services provided it meets the requirements of paragraphs 5.3 to 5.6 of the procurement regulations.

36. **The World Bank has assessed the country procurement regulation and found that the principles and most of the procedures comply with World Bank standards for procurement.** In general, the country's new procurement procedures are in accordance with the World Bank guidelines, but in cases of non-compliance, the provisions of the World Bank prevail. For national competitive bidding procedures for goods and works to become acceptable to the World Bank, some special provisions will be required regarding advertisement, access for foreign bidders to participate, limitation of domestic preference, deadlines for submission of bids, the evaluation and contract award process, standard bidding documents, fraud and corruption clauses in bidding documents, inspection by the World Bank, and obstructive practices and debarment under the national system.

37. The thresholds for market approaches and procurement methods and the World Bank's prior review requirements are indicated in Table 2.6.



Table 2.6: Thresholds for Procurement Methods and Prior Review

No.	Expenditure category	Contract value Threshold* (US\$ equivalent)	Procurement method	Contracts subject to prior review/(US\$ equivalent)
1	Projects	≥15,000,000	Open competition international market approach and direct contracting	≥10,000,000
		500,000–15,000,000	Open competition national market approach	
		≤500,000	Request for quotation	None
2	Goods, information technology, and non-consulting services	≥3,000,000	Open competition international market approach and direct contracting	≥1,500,000
		300,000–3,000,000	Open competition national market approach	
		≤300,000	Request for quotation	None
3	National short list for selection of consultant firms	<400,000	For consulting services	None
		≤500,000	For engineering and construction supervision	None
4	International short list for selection of consultant firms	≥400,000	For consulting services	≥1,000,000
		≥500,000	For engineering and construction supervision	≥1,000,000
5	Selection of individual consultants	All values	All approaches	≥300,000
6	Direct contracting	All values	—	As agreed in procurement plan
7	Training, workshops and study tours	All values	Based on approved annual work plan and budgets	Annual work plan and budgets

38. **Training, workshops, study tours, and conferences.** These activities would comprise workshops and training, based on individual needs; group requirements; on-the-job training; and hiring of consultants to develop training materials and conduct training. Selection of consultants for training services would follow the requirements for selection of consultants mentioned above. All training and workshop activities (other than consulting services) would be conducted on the basis of approved annual work plans and training plans that would identify the general framework of training activities for the year, including type of training or workshop; personnel to be trained; institutions that would conduct training and reason for selection of those institution; justification for the training, how it would lead to effective performance, and implementation of the project; duration of proposed training; and cost estimate for the training.
39. A detailed training and workshop plan indicating the nature of training or workshop, number of trainees or participants, duration, staff months, timing, and estimated cost will be submitted to the World Bank for review and approval before the process is initiated. The selection methods will derive from the activity requirement, schedule, and circumstance. After the training, the beneficiaries will be asked to submit a brief report indicating what skills they have acquired and how these skills will enhance their performance and contribute to attaining the project objective. This report, plus a completion certificate or diploma upon completion of training, shall be provided to the project coordinator and will be kept as part of the record and shared with the World Bank, if required.



40. **Operational costs.** Operational costs that the project finances would be incremental expenses that the PMIA incurs because of project implementation, including in-country travel; office materials and supplies (paper and other consumables, but not limited to the purchase of equipment); office rentals and maintenance; utilities (including electricity and water); communication costs (including telephone and internet charges); equipment rental, operation, maintenance, and repair; and transport costs of the staff related to supervision activities in the field (including per diem). Such services' needs will be procured using the procurement procedures specified in the operations manual, which the World Bank has accepted and approved.
41. **Contract management and expenditure reports.** As part of the procurement management reports, the PMIA will submit contract management and expenditure information quarterly to the World Bank. The procurement management report will contain information on procurement of goods, works, and consultant services and compliance with agreed-upon procurement methods. The report will compare procurement performance against the plan agreed upon at negotiations and as appropriately updated at the end of each quarter. The report will also provide information on any complaints by bidders, unsatisfactory performance by contractors, and contractual disputes and details on payments under each contract. They will be used to ensure that no contract overpayments are made and that not payments are made to sanctioned entities.
42. **The project contracting specialist will ensure that technical managers use STEP during contract management activities to verify strict application of contractual clauses and collect information on all performance indicators.**
43. **Oversight and monitoring arrangements for procurement.** The PIM, which will define the project's internal organization and implementation procedures and include all relevant procedures for calling for bids, selecting consultants, and awarding contracts, will be submitted to the World Bank for review. The project monitoring arrangements for procurement will be specified. The detailed 18-month procurement plan agreed upon with the recipient will be uploaded to the World Bank website. The procurement manual, which may be a section of the PIM, will provide detail on procurement arrangements, roles and responsibilities, methods, and requirements for conducting procurement. The recipient's adoption of the PIM is an effectiveness condition of the financing agreement.
44. **Procurement supervision.** In addition to the prior review and implementation support mission that the World Bank will perform, it is recommended that at least two missions be conducted each year, with one visit to the field to perform a post-review of procurement actions.
45. **Procurement post-review.** World Bank staff; consultants that the World Bank hires; and third parties such as supreme audit institutions, procurement regulatory authorities, consultancy firms, and nongovernmental organizations may perform post-reviews, according to procedures acceptable to the World Bank to ascertain compliance with procurement procedures defined in the legal documents. The sampling is risk based and considers the project procurement risk rating, with riskier projects having a larger sample, and the contract risk rating, to ensure that riskier contracts constitute a greater proportion of the sample. Post-reviews contribute to the overall procurement performance rating of the project and provide a basis for updating project procurement risk and the risk mitigation plan.



ANNEX 3: DETAILED PROJECT DESCRIPTION

46. The proposed project is structured around four components, based on identification of investments, most of which have benefited from detailed studies. The project followed a rigorous selection process with key criteria such as maximum number of people to be provided with access, cities vulnerable to climate change (drought/flood), level of water service, readiness of technical studies for the centers, and security situation as appropriate. For secondary centers, alternative centers are identified to anticipate possible security challenges that may hinder works implementation. The project components are as follows:

47. Component 1: Water supply in Bamako and selected secondary centers (US\$87 million equivalent): This component will ensure safe access to drinking water in the face of climate change—exacerbated droughts and floods by supporting development of efficient, well-performing drinking water services in Bamako and the selected secondary towns (Mopti, San, Sévaré). The component will leverage technical studies²⁹ already conducted with the support of the French Development Agency in Bamako and the secondary centers. Special attention will be paid to the most vulnerable and underserved neighborhoods on the left bank of Bamako, which are also most exposed to floods and droughts. Activities under this component will be phased, initially targeting development of drinking water services in urban areas for which technical studies are already available. In addition to infrastructure investments, technical assistance will be provided to the Malian Water Assets Holding Company (*Société Malienne de Patrimoine de l'Eau Potable*, SOMAPEP-SA) and the Malian Drinking Water Management Company (*Société Malienne de Patrimoine de l'Eau Potable*, SOMAGEP-SA) to improve the overall performance of the sector, in particular its financial viability. These water supply systems will improve the quality of water services by bringing a consistent, drought-resilient water source to water-scarce areas. The systems will also enhance flood resilience by ensuring that installed infrastructure is sited outside flood zones and areas prone to landslides and heatwave and wildfire resilience by using materials that can withstand extreme temperatures.²⁴ Solar power will be used wherever possible as a first option for treatment and pumping energy needs to reduce related greenhouse gas (GHG) emissions and avoid carbon lock-in. When solar is not feasible to cover all energy needs, grid connections will be prioritized over diesel generators or other fossil fuel sources, noting that the electric grid is subject to interruptions. Component 1 will also lower the amount of non-revenue water (NRW), and facilitate the operation of water systems, reducing energy consumption and thus GHG emissions and thereby increasing sector revenues. It will be structured as follows: water supply systems in secondary centers (US\$47 million), water supply system in Bamako (US\$28 million), and management of NRW (US\$12 million). **48. Subcomponent 1.1: Water supply systems in secondary centers (US\$47 million):** This subcomponent will enhance drought resilience by constructing water supply systems in the selected secondary centers to increase access to water supply services. Works include construction of boreholes for production of 24,400 m³/day, three pumping stations, four elevated 2000-m³ water tanks and one 1,000-m³ ground-level water tank, the water distribution network, water transmission pipes, and social household connections and community standposts. SOMAPEP-SA will implement this subcomponent, which will include implementation of protection zones, including necessary legal constraints, upstream of the water production boreholes in the targeted centers, according to article 24 of the 2002 Water Act. If the security situation deteriorates, the project will assess the possibility of relocating to alternative secondary cities. **49. Subcomponent 1.2: Water supply systems in Bamako (US\$28 million):** This subcomponent will construct and supervise works of water supply systems in Bamako, mainly a 9-km transfer from Djikoronni Para to Missira, a pumping station at Missira, a 6-km transfer from Missira to Kati-Sud, a 2,000-m³ elevated water tank, the distribution network, and household connections, to increase drought resilience by increasing access to water services. SOMAPEP-SA will implement this subcomponent. Because the African Development Bank is supporting the transfer of water from Baco Djikoronni to Missira, it will be important to ensure synergy of action with the African Development Bank. This subcomponent will also include exploratory activities to anticipate the need for diversification of

²⁴ In compliance with World Bank [resilient water infrastructure design brief](#).



water sources, such as underground water to further increase the resilience of the system and beneficiaries to climate-exacerbated droughts.

50. Subcomponent 1.3: Management of NRW (US\$12 million): This subcomponent will enhance drought resilience by improving the quality of water supply services, reducing energy use through network loss reduction, improving metering for households and large consumers (companies, administration), procuring equipment to ease leak detection, replacing poor segments of the network, increasing capacity to map networks, and establishing district metering areas. The target for NRW is 5 percent, which corresponds to more than 24,000 m³/day for Bamako alone. Establishment of district metering areas will help manage water pressure to prevent contaminants, such as pathogens or chemicals that can enter through cracks in the system, especially in areas with intermittent supply. Managing pressure reduces outbreak of waterborne diseases by maintaining pipe integrity, minimizing leaks, and protecting water quality. This subcomponent will also help National Directorate of Hydraulics (*Direction Nationale de l'Hydraulique*, DNH), SOMAGEP-SA, and SOMAPEP-SA to improve their data-sharing system on water abstraction and water quality. This will increase detection of overabstraction, flood-related turbidity, and other supply challenges, increasing drought and flood resilience. The subcomponent will also help SOMAGEP-SA improve the operability of water supply systems and support climate change mitigation through NRW, which will reduce GHG emissions and energy costs.

51. Activities under this subcomponent include network rehabilitation works to replace 100 km of poor segments in Bamako and three secondary cities (Bafoulabé/Mahina, Dioila, Ouélessébougou); implementation of a district metering area in Bamako for efficient metering and water supply network management; renewal of household and distribution meters for Bamako, Bafoulabé/Mahina, Dioila, and Ouélessébougou; and installation of production meters in Bafoulabé/Mahina, Dioila, and Ouélessébougou.

52. Bamako's water supply system generates almost 60 percent of the utility's revenues and has the least-efficient water supply network because of the increase in production and water pressure that affect the old network on the left bank of the Niger River. It is estimated that 38 to 40 percent of the water is lost. Bafoulabé/Mahina, Dioila, and Ouélessébougou were transferred to the utility's perimeter in 2018 without any support to update their systems to urban standards. The planned actions will increase network efficiency and metering of the water produced and distributed to consumers.

53. Activities to be financed include technical assistance to SOMAPEP-SA and SOMAGEP-SA to enable the sector to achieve financial viability and improve technical and commercial performance, including customer management over the entire urban perimeter (Bamako and secondary centers). This technical assistance will build on the Utility of the Future tool that the World Bank developed, which begins with an assessment of utility performance and service provision that yields a 100-day action plan for the service provider to implement. Discussions with SOMAPEP-SA and SOMAGEP-SA are ongoing for development of these action plans. The project could then support implementation of this action plan and associated measures through this subcomponent. This subcomponent will support SOMAGEP-SA in Bamako, Bafoulabé-Mahina, Dioila, and Ouélessébougou.

54. Component 2: Enhancing conservation of water sources (US\$8 million equivalent): The infrastructure investments under Component 1 are critically interconnected with the upstream and downstream aspects of the water resources they use, and the changing precipitation regime affects them greatly. For these water sources, this component will support comprehensive water conservation activities to ensure the maintenance and enhancement of ecosystem services that protect against flooding and erosion and allow for enhanced natural aquifer recharge (generating a drought-resilient water supply). These efforts will build on the main institutional water resources management functions at the national and local



levels as outlined in the second phase of the National Program for Integrated Water Resources Management (2023–2026). Key activities will include enhanced climate-resilient planning regarding current and future uses of these water sources and supporting DNH and other institutions by supplying data and tools for secure water resource management. Focusing on the cities targeted under Component 1, this component will increase DNH’s understanding of water resources and their competitive uses and the durability of this resource in times of water scarcity induced by climate change. This includes strengthening the monitoring network for water resource quantity and quality, including at raw water production sites, in part to improve flood and drought early warning systems, and effectively communicating the results of this monitoring to decision-makers and the public. A dialogue will be initiated to begin the journey toward establishing a financially sustainable water resource monitoring solution. By emphasizing consolidation of knowledge and building consensus around resource availability, the project aims to highlight opportunities linked to investments at the sub-basin scale, encompassing groundwater and surface water. This approach will enable DNH to engage actively in dialogues regarding international rivers and transboundary aquifers. The project will reinforce water allocation tools on the Niger River in collaboration with the Niger Basin Authority, ensuring effective management and equitable distribution of these vital resources in the face of greater water variability due to climate change.

55. This component will address questions of turbidity in Niger River water tapped for Bamako and Mopti and in Senegal River water tapped for Bafoulabé and Mahina. A comprehensive evaluation of these basin-level dynamics will be conducted during project implementation, involving thorough dialogue and coordination with key stakeholders, particularly local water committees. To mitigate the impact of solid transport in the rivers, this component will implement nature-based solutions (NBSs) to help stabilize soil in selected areas and riverbanks (which will also increase flood resilience by reducing erosion potential), in close cooperation with the Mali Landscape Restoration Project (P177041). Activities will be conducted in collaboration with local communes and small mining operators working within the riverbeds.

56. For water abstraction facilities tapping groundwater resources in the Airport Zone, Kati, and Dogoudouma for Bamako and in San and Ouélessébougou, this component will establish and implement protection zones, which will include establishing immediate protection areas within which the focus will be on minimizing risks of contamination of the water well.²⁵ Project activities will concentrate on measures to safeguard the physical integrity of wells and prevent pollutants from contaminating the abstracted water. Larger than immediate protection areas, close protection areas are broader areas where climate change, especially heavy downpours, threaten groundwater quality and consequently public health.²⁶ In these areas, efforts will be directed toward minimizing risk of point or accidental contamination of groundwater. In addition to the protection zones, this component will focus on preserving and enhancing natural recharge of aquifers experiencing adverse impacts of changing rainfall patterns by restoring landscapes in the aquifer recharge areas and using NBSs, such as reforestation, construction of terraces, and use of contour bunds, which will sequester carbon, helping mitigate climate change. These targeted approaches will ensure sustainable management and protection of vital groundwater resources, contributing to long-term drought-resilient water security for the communities that these facilities serve.

57. Component 3: Project management and capacity reinforcement (provisional assessment US\$5 million equivalent).

This component will support activities including non-consulting and consulting services; operating costs and training, safeguard activities; monitoring and evaluation; citizen engagement activities; capacity-building activities, including those

²⁵ U.S. Environmental Protection Agency. “Climate Adaptation and Source Water Impacts.” U.S. Environmental Protection Agency, Washington, DC.
<https://www.epa.gov/arc-x/climate-adaptation-and-source-water-impacts#:~:text=Climate%20change%20threatens%20the%20quality,increase%20due%20to%20climate%20change.>

²⁶ U.S. Environmental Protection Agency. “Climate Adaptation and Source Water Impacts.” U.S. Environmental Protection Agency, Washington, DC.
<https://www.epa.gov/arc-x/climate-adaptation-and-source-water-impacts#:~:text=Climate%20change%20threatens%20the%20quality,increase%20due%20to%20climate%20change.>



relevant for the water sector assessment and dialogue with counterpart for sector financial viability, implementation support such as field missions and data collection, and outreach and communication. It will also help finance the project resettlement action plan and support the studies required to prepare new operations.

58. Component 4: Contingency emergency response component (US\$0 million). This zero-funded component will allow the government of Mali to mobilize funds quickly in the event of an emergency that requires immediate recovery or reconstruction response. In the event of a crisis or disaster caused by a natural hazard, including climate-related emergencies such as severe drought and flood, this component enables the government to reallocate International Development Agency project funds to disaster response and recovery purposes under streamlined procedures, thereby supporting Mali's emergency preparedness and response capacity, including financing of critical emergency goods or emergency recovery and associated services, as well as targeted provision of post-disaster support to affected households and individuals. After an adverse natural event, the government's declaration of disaster in accordance with national law, and adherence to the World Bank's activation policy, the contingency emergency response component would be triggered.



ANNEX 4: MAP OF THE PROJECT AREA (Approved on September 17th, 2024) – MLI48362





ANNEX 5: CLIMATE RESILIENCE AND ADAPTATION

1. This annex describes the contribution of project activities to climate adaptation and mitigation. Table 5.1 shows greenhouse gas emissions associated with each subcomponent. Table 5.2 provides the details of adaptation and mitigation-related benefits from each subcomponent and activity, including a cost breakdown.

Table 5.1 : Greenhouse Gas Emissions According to Subcomponent

GHG Emissions at sub-component level								
Components		Totals by Category	Total CO2 Equivalent (tonnes CO2eq) Project Gross Emissions	Total CO2 Equivalent (tonnes CO2eq) Baseline Emissions	Total CO2 Equivalent (tonnes CO2eq) Net Emissions	Total CO2 Equivalent (tonnes CO2eq/year) Net Average Annual Emissions	Percentage reduction over baseline	Substantiality met (Yes/No)
Sub-component 1.1	Boreholes in san + mopti	Emissions for Water Utilities	20573.0	28075.1	-7502.1	-250.1	-26.72	Yes
Sub component 1.2	Pumping stations		6380.0	0.0	6380.0	213.0	0.00	No
Sub-component 1.3	NRW Bamako	Emissions for Water Utilities	879447.2	1420391.4	-540944.2	-18031.5	-38.08	Yes
Sub component 2	FAO Exact (reforestation)	Emissions	-589432	-69510.0	-519922.0	-17330.7	-747.98	Yes
	Total		316968	1378956.6	-1061988.321	-35,400	-77.01	Yes

Table 5.2: Climate Adaptation and Mitigation Contribution According to Activity

Components, subcomponents, and activities	Climate adaptation activities and investments	Climate mitigation activities and investments
Component 1: Water supply in Bamako and selected secondary centers (US\$87 million)		
Subcomponent 1.1: Water supply systems in secondary centers (US\$47 million) - Consultant service works supervision (US\$2.1 million) - Construction of boreholes (US\$0.9 million) - Water supply system, including solar panels, network, water tanks, pumps (US\$ 40.0 millions) - Household connections (US\$2.8 million) - Standposts (US\$1,2 million)	This subcomponent will significantly increase drought resilience in Mali's secondary cities by constructing water supply systems, which will provide reliable, convenient water sources to highly drought-vulnerable populations who did not previously have secure water sources. Because climate change is a major driver of the investments, groundwater and surface water sources, which have different climate vulnerability levels depending on location, will be used to provide multiple water sources, making the systems even more drought resilient. System beneficiaries will no longer have to walk long distances or purchase expensive carted water during times of water scarcity. Infrastructure will	Activities under Subcomponent 1.1 will reduce GHG emissions. The subcomponent will finance construction of boreholes, which will lead to a net reduction of 7,502 tons of carbon dioxide equivalent over lifetime. This subcomponent will reduce emissions by 27% for the new boreholes constructed and by 41% for the water supply system annually from baseline. The project activities described above would primarily rely on solar energy as a primary energy source, which will reduce emissions and promote a low-carbon development pathway. When solar



	be built with resilient design measures to withstand extreme heat and be resilient to floods, and critical infrastructure will be placed outside flood zones.	is not feasible, the electric grid will be used, with diesel used as a backup in emergencies. Storage tanks constructed will use gravity as their source of energy.
Subcomponent 1.2: Water supply systems in Bamako (US\$28 million) - Consultant service works supervision (US\$1.9 million) - Meters to count for water (US\$1.2 million) - Transmission pipes, network, reservoir and pumping station in Bamako US\$ 24.9 millions	This subcomponent will increase drought resilience in Bamako by construction water supply systems, providing the vulnerable urban population with a more-drought-resilient water supply system that will provide convenient, secure access to water during water-scarce periods. The exploratory activities are designed to increase diversity of water supply systems, thereby increasing drought resilience if certain sources become unavailable during droughts. Infrastructure will be built with design measures to withstand extreme heat and be resilient to floods. Expanding the water supply system will enhance resilience by reducing the time needed and physical burden of collecting water, especially for women and children, who walk 2 kilometers on average to fetch water. This will ensure reliable access to water, even during droughts, reducing vulnerability to climate-induced water scarcity.	
Subcomponent 1.3: Management of NRW (US\$12 million) - Consultant service works supervision (US\$0.6 million) - Equipment to detect leaks, network management, pressure management, establishment of district metering areas (US\$6.2 million) - Civil works to repair or replace meters and faulty pipes (US\$ 5.2 million)	This subcomponent will support drought resilience by monitoring the water system, including household and production meters, which will reduce NRW, ensuring less water loss between source and end points, meaning less total water will be pumped and thus less strain on drought-vulnerable water sources. Water source monitoring will allow for early warning of developing drought conditions or turbid floodwaters, so that actions can be taken to prevent severe impacts of floods and droughts on water systems. Establishment of district metering areas will help manage water pressure to prevent contaminants, such as pathogens	Significant GHG emissions will be achieved through NRW reduction in Bamako that will lead to a net reduction of 540,944 tons of carbon dioxide equivalent over project lifetime and 18031 tons of carbon dioxide equivalent annually (a 38% annual reduction from baseline). The procured equipment, including meters, will be used for leak detection, replacement of segments of the network in poor condition, capacity reinforcement of network mapping, and establishment of district metering areas, which will increase the water



	and chemicals, from entering through cracks in the system, especially in areas with intermittent supply. Managing pressure reduces outbreaks of waterborne diseases by maintaining pipe integrity, minimizing leaks, and protecting water quality.	and energy use efficiency of the system, lowering GHG emissions.
Component 2: Increasing conservation of water sources (US\$8 million)		
• Preparatory studies of riverbanks (US\$0.4 million) (adaptation) • Preparatory studies of aquifer recharge areas (US\$0.8 million) (adaptation and mitigation) • Civil works for NBSs for riverbank stabilization (US\$2.9 million) (adaptation) • Civil works for NBSs for soil stabilization and reforestation (US\$2.8 million) (adaptation and mitigation) • Support to local water committees for long-term monitoring and maintenance of riverbanks, soil, and forests (US\$0.5 million; US\$0.3 for riverbanks [adaptation], US\$0.2 for soil and forests [adaptation and mitigation]) • Technical equipment for National Directorate of Hydraulics to monitor soil and water preservation measures (US\$0.3 million) (adaptation and mitigation) Not included (not an adaptation measure): Preparatory studies to protect groundwater quality upstream of well fields (US\$0.3 million)	NBS activities in this subcomponent, including reforestation, soil and riverbank stabilization, and establishment of aquifer protection zones, will enhance natural aquifer recharge, which changing precipitation has severely decreased, by increasing vital water storage, which will increase drought resilience while reducing runoff and erosion, decreasing flood peaks and landslides, and increasing flood resilience. Climate-resilient planning, better water monitoring, and early warning systems will enhance drought and flood preparedness of governments and communities.	Based on a preparatory study conducted to identify optimal soil stabilization and reforestation solutions, reforestation activities under this component will lead to sequestration of 519922 tons of carbon dioxide equivalent over the lifetime of the project. Long-term maintenance that local water committees will perform will lead to sequestration of 17,330 tons of carbon dioxide equivalent annually.

Note: GHG, greenhouse gas; NRW, non-revenue water.



ANNEX 6: COST TABLE

Mali Water Security Project (PASEMa)

PASEMa Project Activities	West African African Financial Community francs	U.S. dollars
Total PASEMa	58,000,000,000	100,000,000
Component 1: Water supply investment	49,760,000,000	85,793,000
Subcomponent 1.1: Water supply systems in secondary centers	26,560,000,000	45,793,000
CONSULTANT SERVICES	1,225,000,000	2,112,000
Control and supervision of drilling works in San and Mopti/Sévaré	34,000,000	59,000
Control and supervision of water storage, transmission, and distribution network works in San and Mopti/Sévaré	1,191,000,000	2,053,000
GOODS	95,000,000	164,000
Acquisition of office furniture, computers and office equipment for engineering assistants, accountants, ES screening and the project management unit	95,000,000	164,000
WORKS	25,240,000,000	43,517,000
Construction of boreholes in San and Mopti/Sévaré	500,000,000	862,000
Drilling equipment, storage facilities, and supply and installation of distribution networks (4 lots)	23,140,000,000	39,897,000
Realization of 20,000 social connections on the distribution network in secondary centers and Bamako (3 lots)	1,600,000,000	2,759,000
o Subcomponent 1.2: Water supply system in Bamako	16,240,000,000	28,000,000
CONSULTANT SERVICES	1,122,984,085	1,936,000
Control and supervision services for water storage, transmission, and distribution works in Bamako	1,122,984,085	1,936,000
GOODS	700,000,000	1,207,000
Supply of 20,000 meter (2 lots)	700,000,000	1,207,000
WORKS	14,417,015,913	24,857,000
Construction of Missira to Kati-Sud transfer pipe, water tower, and pumping station	4,814,138,223	8,300,000
Realization of a transmission pipe to supply Missira reservoirs from Djicoroni-Para station	8,542,800,002	14,729,000
Works for realization of water distribution network (main, secondary, tertiary pipes)	1,060,077,688	1,828,000
Subcomponent 1.3: Management of non-revenue water	6,960,000,000	12,000,000



CONSULTANT SERVICE	330,000,000	569,000
Topographic studies to determine contours in areas not covered by the geographic information system network geographic information system and to update pipe plans	80,000,000	138,000
Consultant service (firm) to collect information for updating SIC and geographic information system databases in Bamako	100,000,000	172,000
Consultant service for civil works supervision for distribution network replacement in Bamako, Bafoulabé/Mahina, Dioila, and Ouélessébougou	150,000,000	259,000
GOODS	3,587,179,056	6,185,000
Acquisition of meters for household connections and production and distribution facilities	1,076,410,386	1,856,000
Acquisition of hydraulic parts for meter installation	225,268,670	388,000
Acquisition of pressure regulation equipment	285,000,000	491,000
Acquisition of distribution network parts for Bamako	195,000,000	336,000
Acquisition of remote management equipment	525,500,000	906,000
Acquisition of meters equipped with remote reading transmitters	200,000,000	345,000
Acquisition of geo-radar and differential global positioning system	250,000,000	431,000
Acquisition of five leak-detection vehicles	490,000,000	845,000
Acquisition of wheeled hydraulic shovel	120,000,000	207,000
Acquisition of truck crane	220,000,000	379,000
WORKS	3,042,820,942	5,246,000
Installation of meters and accessories	168,518,515	291,000
Construction of access points for meters, flowmeters, and pressure regulators	500,502,428	863,000
Work on distribution networks to replace fictitious pipes in of Bamako, Bafoulabé/Mahina, Dioila, and Ouélessébougou (5 lots)	1,623,799,999	2,800,000
Renewal of 25,000 existing household connections (5 lots)	750,000,000	1,293,000
Component 2 : Enhancing conservation of water sources	4,640,000,000	8,000,000
CONSULTING SERVICES	1,364,000,000	2,352,000
Supervision and monitoring of six exploratory boreholes and six piezometers	50,000,000	86,000
Hydrogeological synthesis and identification of water abstraction categories in project area	350,000,000	603,000
Control and monitoring services for development of well fields, repair of degraded riverbanks, and restoration of mined sites in Niger River	130,000,000	224,000
Identification and characterization of aquifer recharge areas	50,000,000	86,000
Control and monitoring services for protection of wellfields and recharge zones in Kabala, San, Mopti, Bafoulabé, Ouélessébougou and Mahina	140,000,000	241,000
Preliminary and detailed designs for construction of groundwater recharge structures (e.g., hillside reservoirs, filtering dikes, underground dams)	100,000,000	172,000



Master plan for water development and management for Upper Niger River, including flow modeling to optimize resource allocation at Bamako and Mopti	325,000,000	560,000
Control and supervision services for installation of specialized equipment for water quality monitoring and surveillance network	30,000,000	52,000
Catering service provider for organization of water resource allocation monitoring meetings	75,000,000	129,000
Technical assistance and support engineer at National Directorate of Hydraulics	54,000,000	93,000
Water quality analysis	60,000,000	103,000
GOODS		328,000
Procurement and installation of five hydrometric stations, five radar flowmeters, and six automatic data recorders		172,000
Procurement of equipment and field missions sets hand piezometer probe (light signal, 200 m with bottom sensor and transport bag) + multiparameter probe: pH, temperature, conductivity, oxidation reduction, oxygen, turbidity; inspection camera		86,000
Procurement of specialized software for cartography, modeling, and remote sensing		69,000
WORKS	3,085,999,998	5,321,000
Construction of six exploratory boreholes and six piezometers	400,000,000	690,000
Rehabilitation of degraded riverbanks and restoration of mined sites in Niger River (2 lots)	1,120,000,000	1,931,000
Wellfield layout and immediate protection perimeter and preservation of aquifer recharge zones in Bafoulabé, Kabala, Mahina, Mopti, Ouélesébougou, and San (3 lots)	1,170,000,000	2,017,000
Installation of specialized equipment for water quality monitoring network	295,999,998	510,000
Development and reforestation of riverbanks in accordance with action plan of the Kangaba, Sansandingue, and Selingue local water committees.	100,000,000	172,000
Component 3: Project management and capacity reinforcement	3,600,000,000	6,207,000
Malian Potable Water Heritage Company	3,259,000,000	5,619,000
Recruitment of support engineer based in San	90,000,000	155,000
Recruitment of support engineer based in Mopti/Sévaré	90,000,000	155,000
Recruitment of accounting assistant to support project financial management specialist	120,000,000	207,000
Recruitment of procurement assistant to support project procurement specialist	120,000,000	207,000
Services of environmental specialist	240,000,000	414,000
Services of social safeguard specialist	150,000,000	259,000
Services of four junior environmental and social experts	45,000,000	78,000



Acquisition of office furniture, computers, and office equipment for engineering assistants, accountants, Environmental and social safeguards, and project management unit	95,000,000	164,000
Recruitment of financial and accounting audit consultant	100,000,000	172,000
Support for grievance management committees	24,000,000	41,000
Recruitment of nongovernmental organization to provide citizen engagement services to people in Bamako, Mopti/Sévaré, and San	235,000,000	405,000
Consultant service for elaboration environmental and social impact assessment and resettlement action plan of the water transmission pipe from the Djicoroni station to Missira	75,000,000	129,000
Implementation of project resettlement action plans	900,000,000	1,552,000
Consultant service for project implementation completion report	50,000,000	86,000
Consultant service for technical audit of project investments	100,000,000	172,000
Strengthening of Malian Potable Water Heritage Company information technology system (e.g., management software, mapping software, backup software, storage servers)	150,000,000	259,000
Client satisfaction survey	75,000,000	129,000
Payment of travel expenses for project follow-up (e.g., fuel, per diem, tolls)	250,000,000	431,000
Study tour and benchmarking	200,000,000	345,000
Project communication	50,000,000	86,000
Organizational and institutional studies	100,000,000	172,000
Malian Drinking Water Management Company	121,000,000	209,000
Payment of travel expenses for project follow-up (e.g., fuel, per diem, tolls)	51,000,000	88,000
Study tour and benchmarking	70,000,000	121,000
National Directorate of Hydraulics	220,000,000	379,000
Procurement of computer equipment for technical assistance and support engineer	5,000,000	9,000
Procurement of furniture for technical assistance and /support engineer	5,000,000	9,000
Procurement of 4x4 vehicle	35,000,000	60,000
Travel expenses for project follow-up (e.g., fuel, per diem, tolls, vehicle repairs, taxes, insurance).	75,000,000	129,000
Study tour and benchmarking	100,000,000	172,000
Component 4: Contingency emergency response component		